MultiHU-TD: Multifeature Hyperspectral Unmixing Based on Tensor Decomposition

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***Abstract*— Hyperspectral unmixing allows representing mixed pixels as a set of pure materials weighted by their abundances. Spectral features alone are often insufficient, so it is common to rely on other features of the scene. Matrix models become insufficient when the hyperspectral image (HSI) is represented as a high-order tensor with additional features in a multimodal, multifeature framework. Tensor models such as canonical polyadic decomposition allow for this kind of unmixing but lack a general framework and interpretability of the results. In this article, we propose an interpretable methodological framework for low-rank multifeature hyperspectral unmixing based on tensor decomposition (MultiHU-TD) that incorporates the abundance sum-to-one constraint in the alternating optimization alternating direction method of multipliers (ADMM) algorithm and provide in-depth mathematical, physical, and graphical interpretation and connections with the extended linear mixing model. As additional features, we propose to incorporate mathematical morphology and reframe a previous work on neighborhood patches within MultiHU-TD. Experiments on real HSIs showcase the interpretability of the model and the analysis of the results. `Python` and `MATLAB` implementations are made available on GitHub.**

***Index Terms*— Blind source separation (BSS), extended linear mixing model (ELMM), hyperspectral unmixing (HU), interpretability, tensor decomposition.**

NOMENCLATURE

A. Matrix and Tensor Decomposition

ADMM	Alternating direction method of multipliers.
ALS	Alternating least squares.
AO	Alternating optimization.
AO-ADMM	Alternating optimization ADMM.
ASC	Abundance sum-to-one constraint.
BSS	Blind source separation.
BTD	Block term decomposition.
CPD	Canonical polyadic decomposition.
NMF	Nonnegative matrix factorization.
ProCo-ALS	Projected compressed ALS.

B. Hyperspectral Imaging

AM	Abundance map.
ELMM	Extended linear mixing model.
EM	Endmember.
GT	Ground truth.
HSI	Hyperspectral image.
HU	Hyperspectral unmixing.
LiDAR	Light detection and ranging.
LMM	Linear mixing model.
MultiHU-TD	Multifeature hyperspectral unmixing based on tensor decomposition.
RMSE	Root mean square error.
SAD	Spectral angular distance.
SNR	Signal-to-noise ratio.
SV	Spectral variability.

C. Mathematical Morphology

CbR	Closing by reconstruction.
EMP	Extended morphological profile.
MM	Mathematical morphology.
MP	Morphological profile.
ObR	Opening by reconstruction.
SE	Structuring element.

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I. INTRODUCTION

HYPERSPECTRAL IMAGING refers to the acquisition of images of a scene over a wide and almost continuous spectrum. An HSI contains pixels that can cover areas of pure or mixed materials and amounts to a high spectral feature diversity [1], [2]. These characteristics allow to perform BSS [3], [4], [5] on the observed spectral signatures to blindly extract those of pure materials (*sources*), also called EMs, and their per-pixel (*per-sample*) abundances. This case of BSS is known as HU, which is an active research topic with several applications, such as remote sensing, chemometrics, and biomedical imagery [6], [7], [8], [9], [10], [11], [12],

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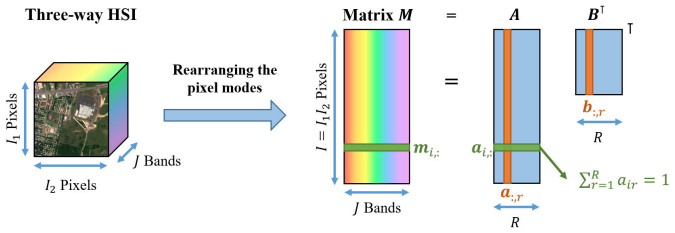


Fig. 1. Classical matrix-based HU using NMF (i.e., LMM).

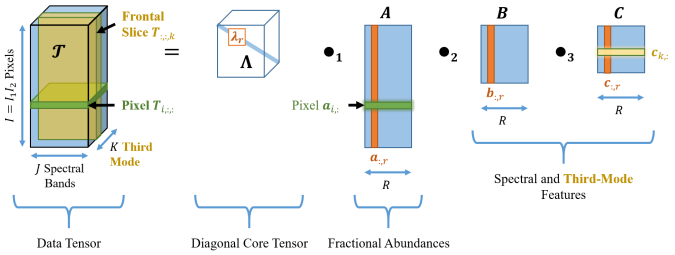


Fig. 2. CPD of a third-order tensor. The tensor is formed of K matricized HSIs that are stacked along the third mode. A frontal slice $\boldsymbol{T}_{:,:,k}$ (in yellow) represents a matricized HSI and is associated with one row of $\boldsymbol{C}$. A horizontal (pixel) slice $\boldsymbol{T}_{i,:,:}$ (in green) represents a matrix of features and is associated with one row of $\boldsymbol{A}$.

[13], [14], [15]. HU allows understanding and quantifying the physical components of a scene.

A significant part of research in BSS and HU relies on matrix factorization with additional constraints that aim at modeling the context of the problem. Consequently, an observed data matrix $\boldsymbol{M} \in \mathbb{R}^{I\times J}$ (i.e., with I pixel samples and J spectral features) is decomposed into two factor matrices $\boldsymbol{A} \in \mathbb{R}^{I\times R}$ and $\boldsymbol{B} \in \mathbb{R}^{J\times R}$ such that

$$\boldsymbol{M} = \boldsymbol{A}\boldsymbol{B}^{\mathsf{T}} = \sum_{r=1}^{R} \boldsymbol{a}_r \boldsymbol{b}_r^{\mathsf{T}} \tag{1}$$

where R is the number of latent components to be estimated, and $\boldsymbol{a}_r$ and $\boldsymbol{b}_r$ are the columns of $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively, $\forall\, r \in \{1, \ldots, R\}$. As such, the columns of $\boldsymbol{B}$ represent the estimated source signals, and the rows of $\boldsymbol{A}$ represent the per-sample abundances of the sources. The decomposition is often carried out by minimizing the generic cost function [16]

$$\underset{\boldsymbol{A},\boldsymbol{B}}{\operatorname{argmin}} \|\boldsymbol{M} - \boldsymbol{A}\boldsymbol{B}^{\mathsf{T}}\|_F^2 + r(\boldsymbol{A}) + r(\boldsymbol{B}) \tag{2}$$

where $r(\cdot)$ encodes the imposed constraints and/or regularizations to enforce desirable properties on the solutions.

In the case of HU, a classical approach is the NMF, which relies on the LMM of the observed HSI matrix (see Fig. 1). Hence, $\boldsymbol{A}$ and $\boldsymbol{B}$ are elementwise *nonnegative*, which applies also in most domains of BSS (other than HU) where the interpretability of the factor matrices is important. Moreover, the rows of $\boldsymbol{A}$ are subject to the ASC, which means that each row sums to 1

$$\sum_{r=1}^{R} a_{ir} = 1 \quad \forall\, i \in \{1, \ldots, I\} \tag{3}$$

which applies to domains where the coefficients of the decomposition are proportions.

When only few materials concur in the mixture for each pixel, sparsity is imposed on the abundances [17]. Finally, real HSIs often contain SVs in the sources, e.g., variations in the EMs due to local physicochemical variations, illumination changes, or topographic effects. In order to account for these SVs, the ELMM was proposed to extend the LMM to account for said SVs, which is an active topic that has seen a lot of progress recently [11], [12], [13], [14].

A. Tensor Analysis of HSIs

An HSI can be treated as a data cube [18], [19], [20], [21], [22], [23] (i.e., a third-order tensor with two spatial and one spectral dimensions). However, sometimes, the HSI does not come alone but is associated with additional modalities such as the following.

1) A time series or multiangular data of HSI images [24].
2) The HSI is combined with images acquired by different sensors (e.g., panchromatic, multispectral, and LiDAR fusion) [25], [26], [27].
3) Some spatial features are extracted from the HSI (such as in spectral–spatial classification problems [28], [29], [30], [31]).

Such scenarios have recently also concerned other areas of BSS such as multichannel signal processing [32], [33], [34] and multidimensional biomedical signal and image processing [35], [36], [37], [38], [39]. In the aforementioned scenarios, the data are represented natively as *tensors*[1] [40], and the challenge usually boils down to the proper modeling of a joint factorization of multivariate representations without losing the multimodal structure and, hence, its interpretation in terms of BSS.

Among these scenarios, we focus on the case of HU where the HSI is associated with an additional set of features in the form of a new tensor mode, which we coin as MultiHU-TD. For example, we consider a set of spatial features extracted from the image itself that can be considered as new modes. Moreover, the pixels are rearranged in lexicographic order, promoting *low-rank* tensor decomposition.[2] In this scenario, there are some challenging questions to answer, such as how can we jointly perform a constrained factorization in such settings? and how can we interpret the extracted factors? An analysis of the literature shows that there are works that perform NMF with additional constraints [41], [42], [43] and others that consider the case of multimodal inputs with coupled NMF [44], [45], but this is different from considering data as tensors in our case.

As the native structure of our data is a tensor, we consider the problem in terms of tensor decomposition [40], which is the natural framework for processing multimodal data in the signal and image processing community [46], [47], [48]. There

[1] A tensor can be represented as a multidimensional array. The order of a tensor refers to the number of its array's indices, which is also the number of its *modes*. For example, a tensor of dimensions $I \times J \times K$ is said to have three *modes* and is called a *third-order tensor*. Datasets with order 3 or above are described as *high-order* tensors.

[2] Pixels form only one mode, even if images are often seen as 2-D objects. This suppresses the high-rankness introduced by the complex spatial features of the whole scene [15], which is an inconvenience for BSS methods [26].

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are many types of decomposition, such as tucker decomposition, BTD, and CPD [49]. However, we choose CPD thanks to the diagonality of its core tensor, which allows the interaction only between similarly indexed columns in the factor matrices, which *naturally* promotes *source separation* and a straightforward *interpretability* of the extracted components in unmixing, both of which are core aspects of our work.

In the third-order case, as illustrated in Fig. 2, CPD decomposes a data tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{I\times J\times K}$ into a diagonal core tensor $\boldsymbol{\Lambda} \in \mathbb{R}^{R\times R\times R}$ and three factor matrices $\{\boldsymbol{A} \in \mathbb{R}^{I\times R}, \boldsymbol{B} \in \mathbb{R}^{J\times R}, \boldsymbol{C} \in \mathbb{R}^{K\times R}\}$, each representing one of the three modes of $\boldsymbol{\mathcal{T}}$, respectively, such that

$$\boldsymbol{\mathcal{T}} = \boldsymbol{\Lambda} \underset{1}{\bullet} \boldsymbol{A} \underset{2}{\bullet} \boldsymbol{B} \underset{3}{\bullet} \boldsymbol{C} \tag{4}$$

where $\bullet_d$ denotes the mode-d product (product along the dth mode), further described in expressions (6), (13b), and (13c).

CPD extends NMF to high-order data and can adopt all of its features, especially that of imposing constraints. CPD is often computed by minimizing the cost function [40]

$$\underset{\boldsymbol{A},\boldsymbol{B},\boldsymbol{C}}{\operatorname{argmin}} \|\boldsymbol{\mathcal{T}} - \boldsymbol{\Lambda} \underset{1}{\bullet} \boldsymbol{A} \underset{2}{\bullet} \boldsymbol{B} \underset{3}{\bullet} \boldsymbol{C}\|_F^2 + r(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}) \tag{5}$$

where $r(\cdot)$ encodes the imposed constraints. Note that the nonnegative constraint ensures the existence of a minimum; in fact, without an appropriate regularization term $r(\cdot)$, the above cost function could admit only an infimum, which may not be reachable [50]. On the other hand, with an appropriate regularization, problem (5) is well-posed.

B. Related Works and Limitations

In the context of MultiHU-TD, CPD has been used with multitemporal/angular HSIs [24] and with HSIs having an additional diversity of extracted neighborhood patches [14]. Moreover, some works [28], [29], [30] jointly considered HSIs with spatial features extracted by MM filters [51], [52], [53], [54] in the framework of multifeature scene classification. These works show that CPD is a suitable approach for joint decomposition. However, they present some limitations.

1) Algorithmic Perspective: The nonnegative constraint is implemented in [55] by projection onto the nonnegative orthant, which exhibits some computational issues [47]. In [14], nonnegative ALS is used where ASC is also naively implemented by projecting the abundances on the unit simplex, contrary to the common practice in the matrix case [17] where ASC is embedded in the updates. In [24], the nonnegative CPD is computed using the ProCo-ALS algorithm, which is considerably fast [46] but not so flexible with additional constraints. Finally, in [28], [29], and [30], an alternative algorithm is proposed based on AO-ADMM [48] with compression and nonnegative constraints, which is flexible and stable with large datasets, but has not yet addressed MultiHU-TD, which requires further modeling (i.e., sparsity and ASC).

2) Interpretability: The work of Veganzones et al. [24] faced a certain challenge in interpreting the third-mode factors, perhaps due to the naive employment of CPD. In [14], a link was established between CPD and ELMM but was not deeply investigated as it was restricted to the case of patches and tested only with synthetic data. Moreover, it faced another challenge in interpreting the factors, which poses an ambiguity on its performance and the meaning of the extracted features. Finally, Veganzones et al. [14] and Jouni et al. [28], [29], [30] perform tensor decomposition with spatial features. However, the used spatial features can allow limited flexibility (e.g., patches), or the interpretation of the factor matrices was not addressed, noting here that incorporating them with CPD showed improvement in supervised classification (e.g., MM).

TABLE I
ARRAY NOTATIONS

Type	Font style	Example
Scalars	unformatted lowercase	a, b, c, t
Vectors	bold lowercase	$\boldsymbol{a}$, $\boldsymbol{b}$, $\boldsymbol{c}$, $\boldsymbol{t}$
Matrices	bold uppercase	$\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, $\boldsymbol{T}$
Tensors	bold calligraphic	$\boldsymbol{\mathcal{T}}$
Dimension	unformatted uppercase	I, J, K, R
Indices	lowercase version of the spanned dimension	i, j, k, r

In this article, we wish to consider such operations in a BSS framework from the lens of MultiHU-TD with in-depth interpretability. This presents us with two main challenges:

1) tuning AO-ADMM to incorporate ASC, which is challenging due to the multilinear structure of CPD, particularly in modeling the samples as a convex combination of the spectral sources in a multimodal setting;
2) exploring the meaning of the extracted features in these conditions.

C. Contributions

To the best of our knowledge, imposing ASC in CPD or AO-ADMM as a natural extension of NMF in [17] has not been done. Furthermore, we are interested in finding a generalized framework for MultiHU-TD favoring the interpretation of its results under any third-mode diversity. More precisely, our contributions to jointly deal with these limitations are given as follows.

1) We propose a methodological framework for dealing with MultiHU-TD based on AO-ADMM by Huang et al. [48] and expand it to incorporate ASC with joint nonnegativity and sparsity. The proposed AO-ADMM-ASC is a general algorithm that can be applied in other domains of BSS where convex combinations of sources apply.
2) We establish a unified framework for the interpretability of MultiHU-TD. In particular, the link between ELMM and CPD [14] is expanded by providing in-depth physical and graphical insights for better interpretability of the CPD model and its factors.
3) We propose to include MM as spatial features to perform a spectral–spatial HU and demonstrate the aforementioned points. We also revise [14] and provide detailed interpretations on the cases of patches and MM, which has not been addressed in any of the previous works [14], [28], [29], [30]. This analysis also shows

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TABLE II

PIXELS ARE REARRANGED IN *Lexicographic Order* SPANNING THE FIRST MODE, SO I IS THE TOTAL NUMBER OF PIXELS. THE SYMBOL ":" IN THE INDEX INDICATES A SPAN OF THE WHOLE MODE. FOR EXAMPLE, $\boldsymbol{m}_{i,:}$ AND $\boldsymbol{m}_{:,j}$ REPRESENT THE iTH ROW AND jTH COLUMN *Vectors* OF $\boldsymbol{M}$, RESPECTIVELY (SEE FIG. 1), AND $\boldsymbol{T}_{:,:,k}$ REPRESENTS THE kTH FRONTAL *matrix* SLICE OF $\boldsymbol{\mathcal{T}}$ (SEE FIG. 2)

Type	Symbol	Dimensions (pixel × band × transform)	i-th pixel	j-th band	k-th transform	(i,j,k)-th element
HSI matrix	$\boldsymbol{M}$	$I \times J$	$\boldsymbol{m}_{i,:}$	$\boldsymbol{m}_{:,j}$	-	$m_{i,j}$
HSI tensor	$\boldsymbol{\mathcal{T}}$	$I \times J \times K$	$\boldsymbol{T}_{i,:,:}$	$\boldsymbol{T}_{:,j,:}$	$\boldsymbol{T}_{:,:,k}$	$t_{i,j,k}$

TABLE III

TENSOR SLICING AND MODE UNFOLDING

Variable	Symbol	Dimensions
Horizontal slice	$\boldsymbol{T}_{i,:,:}$	$J \times K$
Lateral slice	$\boldsymbol{T}_{:,j,:}$	$I \times K$
Frontal slice	$\boldsymbol{T}_{:,:,k}$	$I \times J$
Mode-1 unfolding	$\boldsymbol{T}_{(1)}$	$JK \times I$
Mode-2 unfolding	$\boldsymbol{T}_{(2)}$	$IK \times J$
Mode-3 unfolding	$\boldsymbol{T}_{(3)}$	$IJ \times K$

that MM is better suited since it embeds physically meaningful features (scale and brightness of objects) into HU, unlike patches.

That said, note that our main goal is to reason about the interpretability of this factorization and to describe this framework rather than to propose yet another HU algorithm.

The remainder is organized as follows. In Section III, we introduce some background. In Section IV, we detail the proposed framework. In Section V, we present our experiments and results. Finally, we draw out some conclusions in VI.

II. NOTATIONS AND DEFINITIONS

Table I shows a list of notations for the different types of objects used throughout this article: scalars, vectors, matrices, tensors, and array dimensions and indices. Table II denotes the types of observed data in this article, their dimensions, and different ways of indexing.

Table III denotes the different ways to slice and unfold a third-order tensor. The *mode unfolding* (or *matricization*) of a tensor means to reshape it into a matrix by fixing the targeted mode and rearranging the others in lexicographic order.

Table IV denotes the factor matrices of an NMF (matrix case) or CPD (tensor case). Mode-1, Mode-2, and Mode-3 correspond to the modes of *pixels*, spectral *bands*, and set of extracted spatial features (*transforms*), respectively.

We use the notation "diag$\{\boldsymbol{v}\}$" to refer to the diagonal matrix whose entries are the elements of any vector $\boldsymbol{v}$.

The *outer product* of two vectors $\boldsymbol{a} \in \mathbb{R}^I$ and $\boldsymbol{b} \in \mathbb{R}^J$ results in a matrix $\boldsymbol{M} \in \mathbb{R}^{I \times J}$ as follows:

$$\boldsymbol{M} = \boldsymbol{a} \otimes \boldsymbol{b} = \boldsymbol{a}\boldsymbol{b}^{\mathsf{T}} \Longleftrightarrow m_{i,j} = a_i b_j \quad \forall i \in \{1,\ldots,I\} \quad \forall j \in \{1,\ldots,J\}.$$

The *outer product* of three vectors $\boldsymbol{a} \in \mathbb{R}^I$, $\boldsymbol{b} \in \mathbb{R}^J$, and $\boldsymbol{c} \in \mathbb{R}^K$ results in a third-order tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{I \times J \times K}$ as follows:

$$\boldsymbol{\mathcal{T}} = \boldsymbol{a} \otimes \boldsymbol{b} \otimes \boldsymbol{c} \Longleftrightarrow t_{i,j,k} = a_i b_j c_k \quad \forall i \in \{1,\ldots,I\} \quad \forall j \in \{1,\ldots,J\} \; \forall k \in \{1,\ldots,K\}.$$

The *mode-d product* $\bullet_d$ represents the product of a tensor by a matrix along the dth mode. For example, assuming that we have $\boldsymbol{\mathcal{G}} \in \mathbb{R}^{L \times M \times N}$, $\boldsymbol{A} \in \mathbb{R}^{I \times L}$, and $\boldsymbol{B} \in \mathbb{R}^{J \times M}$, the mode-1 and mode-2 product of $\boldsymbol{\mathcal{G}}$ by $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively, results in a tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{I \times J \times N}$ defined as

$$\boldsymbol{\mathcal{T}} = \boldsymbol{\mathcal{G}} \underset{1}{\bullet} \boldsymbol{A} \underset{2}{\bullet} \boldsymbol{B} \Longleftrightarrow t_{ijn} = \sum_{l=1}^{L} \sum_{m=1}^{M} G_{lmn}\, a_{il} b_{jm}. \tag{6}$$

III. BACKGROUND

In this section, we briefly review the existing notions in the literature upon which we base our algorithm and generalized interpretation of the MultiHU-TD framework. First, we explain how ASC is applied in NMF [17] as the proposed framework extends this for tensor decomposition. Then, we give a brief account of ELMM, including graphical and visual interpretations, which will be the basis for the proposed interpretation. Finally, we discuss the link between CPD and ELMM preliminarily presented in [14].

A. NMF With ASC, Nonnegativity, and Sparsity

In the NMF case [17], when sparsity (ℓ_1 norm) and ASC are imposed on the abundances, (1) becomes

$$\underset{\boldsymbol{A},\boldsymbol{B}}{\operatorname{argmin}} \frac{1}{2}\|\boldsymbol{M} - \boldsymbol{A}\boldsymbol{B}^{\mathsf{T}}\|_F^2 + \alpha\|\boldsymbol{A}\|_1 \quad \text{s.t. } \boldsymbol{A} \succeq 0, \quad \boldsymbol{B} \succeq 0, \quad \sum_{r=1}^{R} a_{i,r} = 1 \;|_{\forall i \in \{1,\ldots,I\}} \tag{7}$$

where $\alpha > 0$ and $\succeq$ denotes elementwise nonnegativity. A simple strategy to embed ASC goes by stacking a row vector in $\boldsymbol{B}$ and a column vector in $\boldsymbol{M}$ such that [17]

$$\tilde{\boldsymbol{M}} = \left[\boldsymbol{M} \mid \delta \mathbf{1}_{I\times 1}\right], \quad \tilde{\boldsymbol{B}} = \begin{bmatrix} \boldsymbol{B} \\ \delta \mathbf{1}_{1\times R} \end{bmatrix} \tag{8}$$

where δ is a constant that is usually set as the mean of $\boldsymbol{M}$ and the last row of $\tilde{\boldsymbol{B}}$ is reset to δ after each iteration. This operation ensures that ASC is softly embedded in NMF since, $\forall i \in \{1,\ldots,I\}$, we have

$$m_{i,J+1} = \sum_{r=1}^{R} a_{i,r} b_{J+1,r} = \sum_{r=1}^{R} a_{i,r}\delta = \delta \tag{9}$$

corresponding to $\sum_{r=1}^{R} a_{i,r} = 1$. Then, (7) becomes

$$\underset{\boldsymbol{A},\tilde{\boldsymbol{B}}}{\operatorname{argmin}} \frac{1}{2}\|\tilde{\boldsymbol{M}} - \boldsymbol{A}\tilde{\boldsymbol{B}}^{\mathsf{T}}\|_F^2 + \alpha\|\boldsymbol{A}\|_1, \quad \text{s.t. } \boldsymbol{A} \succeq 0, \tilde{\boldsymbol{B}} \succeq 0. \tag{10}$$

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TABLE IV
FACTOR MATRICES, EACH CORRESPONDING TO ONE OF THE MATRIX OR TENSOR MODES. R IS THE NUMBER OF RANK-1 ADDITIVE TERMS IN THE DECOMPOSITION

Factor Mat.	Symbol	Dimensions	Row index	Col. index	Element index
Mode-1	$\boldsymbol{A}$	$I \times R$	$\boldsymbol{a}_{i,:}$	$\boldsymbol{a}_{:,r}$	$a_{i,r}$
Mode-2	$\boldsymbol{B}$	$J \times R$	$\boldsymbol{b}_{j,:}$	$\boldsymbol{b}_{:,r}$	$b_{j,r}$
Mode-3	$\boldsymbol{C}$	$K \times R$	$\boldsymbol{c}_{k,:}$	$\boldsymbol{c}_{:,r}$	$c_{k,r}$

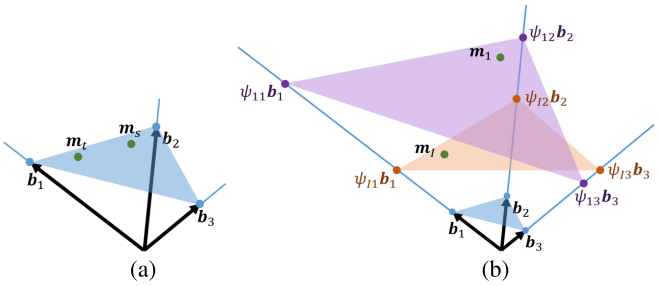


Fig. 3. Graphical comparison between (a) LMM and (b) ELMM (scalar factors) in the case of three spectral signatures $\{\boldsymbol{b}_{:,1}, \boldsymbol{b}_{:,2}, \boldsymbol{b}_{:,3}\}$ and two example pixels $\{\boldsymbol{m}_{s,:}, \boldsymbol{m}_{t,:}\}$. The relative coordinates in the simplices are conserved.

There are many algorithms proposed in the literature that deal with sparse NMF and ASC, which are out of the scope of this work [41], [42]. In our case, we extend NMF within the AO-ADMM framework for CPD. NMF then becomes a special case for order-2 tensors. This tensor extension proposed in Section IV-A is referred to as AO-ADMM-ASC.

B. ELMM

While LMM is seen as a direct approach for HU, it cannot model SVs represented by nonlinear effects or illumination conditions. One way to account for said effects is through ELMM [12], which, in general, assumes additional degrees of freedom that account to said SVs *at the pixel level* by introducing a *pixel-dependent* SV function $\boldsymbol{f}_i : \mathbb{R}^J \to \mathbb{R}^J$ $\forall\, i \in \{1, \ldots, I\}$, which maps each EM $\boldsymbol{b}_{:,r}$ $\forall\, r \in \{1, \ldots, R\}$ to a new spectral signature $\boldsymbol{b}_{:,r}^{(i)}$ that best reflects the targeted SVs

$$\boldsymbol{m}_i = \sum_{r=1}^{R} a_{ir}\, \boldsymbol{f}_i(\boldsymbol{b}_{:,r}) = \sum_{r=1}^{R} a_{ir} \boldsymbol{b}_{:,r}^{(i)}. \tag{11}$$

For example, in the case of different illumination conditions, this can be represented as a scaling factor for each pixel on the EMs. In the following, we present the parts that are at the basis of the interpretability of our proposed framework.

When ASC is imposed in LMM, all the pixels will lie on the convex hull of the set of estimated EMs (i.e., the columns of $\boldsymbol{B}$[3]), and the fractional abundances in each row of $\boldsymbol{A}$ define the coordinates of each pixel on the convex hull, which is illustrated in Fig. 3(a). With the introduction of ELMM, the pixels will not lie on the same simplex anymore as each pixel is mapped to a new set of EMs, which is illustrated in Fig. 3(b) in the case where the SV is modeled by a scaling factor ψ_{ir} that is pixel- and EM-dependent such that $\boldsymbol{b}_r^{(i)} = \psi_{ir} \boldsymbol{b}_{:,r}$ [12]. Accordingly, the new spectral signatures $\boldsymbol{b}_r^{(i)}$ are allowed to move only along the directions of $\boldsymbol{b}_{:,r}$.

In the case where the SVs are modeled with scaling factors, we denote by $\boldsymbol{\Psi} \in \mathbb{R}^{I \times R}$ the matrix whose entries are the scaling factors ψ_{ir} $\forall\, i \in \{1, \ldots, I\}$ and $\forall\, r \in \{1, \ldots, R\}$. Then, the following are equivalent:

$$\boldsymbol{M} = (\boldsymbol{A} \boxdot \boldsymbol{\Psi})\boldsymbol{B}^\mathsf{T} \iff \tag{12a}$$

$$\boldsymbol{m}_{i,:} = \sum_{r=1}^{R} a_{ir} \psi_{ir} \boldsymbol{b}_{:,r}^\mathsf{T} = \boldsymbol{a}_{i,:} \operatorname{diag}\{\boldsymbol{\Psi}_{i,:}\}\boldsymbol{B}^\mathsf{T} = \boldsymbol{a}_{i,:}\boldsymbol{\Psi}_{(i)}\boldsymbol{B}^\mathsf{T} \tag{12b}$$

where $\boxdot$ is the Hadamard product. $\boldsymbol{\Psi}_{i,:} \in \mathbb{R}^R$ is the ith row of $\boldsymbol{\Psi}$, and $\boldsymbol{\Psi}_{(i)} \in \mathbb{R}^{R \times R}$ represents the diagonal matrix formed from $\boldsymbol{\Psi}_{i,:}$. These mathematical, graphical, and visual relationships are the key to elaborate and interpret the case of CPD in Section IV-B.

C. CPD and ELMM

In the work of Veganzones et al. [14], a representation of CPD in terms of ELMM has been presented, which will be reported here. CPD decomposes a third-order tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{I \times J \times K}$ such that

$$\boldsymbol{\mathcal{T}} = \boldsymbol{\Lambda} \underset{1}{\bullet} \boldsymbol{A} \underset{2}{\bullet} \boldsymbol{B} \underset{3}{\bullet} \boldsymbol{C} \Leftrightarrow \boldsymbol{\mathcal{T}} = \sum_{r=1}^{R} \lambda_{r,r,r}\, \boldsymbol{a}_{:,r} \otimes \boldsymbol{b}_{:,r} \otimes \boldsymbol{c}_{:,r} \tag{13a}$$

$$\Leftrightarrow \boldsymbol{\mathcal{T}} = \sum_{r=1}^{R} \boldsymbol{a}_{:,r} \otimes \boldsymbol{b}_{:,r} \otimes \boldsymbol{\psi}_{:,r} \tag{13b}$$

$$\Leftrightarrow t_{i,j,k} = \sum_{r=1}^{R} a_{i,r}\, b_{j,r}\, \psi_{k,r} \tag{13c}$$

where $t_{i,j,k}$ is a (scalar) entry of $\boldsymbol{\mathcal{T}}$, R is the number of estimated sources, and $\boldsymbol{\Lambda} \in \mathbb{R}^{R \times R \times R}$ is a diagonal tensor that absorbs the ℓ_2-norms of the columns of the factor matrices [40], [47]. In (13b) and (13c), we suppress the expression of $\boldsymbol{\Lambda}$ by absorbing its entries in the columns of $\boldsymbol{C}$, resulting in $\boldsymbol{\Psi} \in \mathbb{R}^{K \times R}$ whose columns are the scaled version of those of $\boldsymbol{C}$ such that $\boldsymbol{\psi}_{:,r} = \lambda_{r,r,r} \boldsymbol{c}_{:,r}$ $\forall\, r$.

As shown in [14], given that $\boldsymbol{T}_{:,:,k}$ denotes the kth frontal slice of $\boldsymbol{\mathcal{T}}$, $\forall\, k \in \{1, \ldots, K\}$, and assuming that $\boldsymbol{\Psi}_{(k)} \in \mathbb{R}^{R \times R}$ represents the diagonal matrix formed from the row $\boldsymbol{\psi}_{k,:} \in \mathbb{R}^R$ of the factor $\boldsymbol{\Psi}$, one can write

$$\boldsymbol{T}_{:,:,k} = \boldsymbol{A} \operatorname{diag}\{\boldsymbol{\psi}_{k,:}\}\boldsymbol{B}^\mathsf{T} = \boldsymbol{A}\boldsymbol{\Psi}_{(k)}\boldsymbol{B}^\mathsf{T} = \boldsymbol{A}\tilde{\boldsymbol{f}}_k(\boldsymbol{B})^\mathsf{T} \tag{14a}$$

$$\iff \boldsymbol{t}_{i,:,k} = \sum_{r=1}^{R} a_{i,r}\, (\boldsymbol{b}_{:,r} \psi_{k,r}) = \sum_{r=1}^{R} a_{i,r}\, \boldsymbol{f}_k(\boldsymbol{b}_{:,r}) \tag{14b}$$

[3]Here, we note that, unless the EMs are not affinely independent, which is unlikely, and $R \leq J + 1$, then the convex hull is a simplex.

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where $\boldsymbol{t}_{i,:,k}$ is the ith pixel row of $\boldsymbol{T}_{:,:,k}$. From the perspective of each frontal slice, CPD resembles a regularized coupled NMF of the frontal slices where $\boldsymbol{A}$ is a common factor

$$\underset{\boldsymbol{A},\boldsymbol{B}^{(k)}}{\operatorname{argmin}} \|\boldsymbol{T}_{:,:,k} - \boldsymbol{A}\boldsymbol{B}^{(k)\mathsf{T}}\|_F^2 \text{ s.t. } \boldsymbol{A} \succeq 0, \boldsymbol{B}^{(k)} \succeq 0 \tag{15}$$

where $\boldsymbol{B}^{(k)} = \tilde{\boldsymbol{f}}_k(\boldsymbol{B})|_{\forall k \in \{1,\ldots,K\}}$, which has an analogous expression to the case of ELMM. The latter was used in [14] as a way to impose a spatial smoothing (15) on the abundances within a patch of neighboring pixels. This link between CPD and ELMM was preliminary presented and restricted to an application of patches with only simulated data, and the meaning of the SVs was not explored.

In Section IV-B, we propose a generalized in-depth interpretation of MultiHU-TD, and in Section IV-C, we present how to incorporate spatial features (e.g., patches and MM) in such a setting.

IV. Proposed MultiHU-TD Framework

In this section, we present the proposed MultiHU-TD framework based on CPD. First, we talk about the implementation of AO-ADMM-ASC. Then, we provide an interpretation of MultiHU-TD, including the physical meaning of ASC and the ELMM model in the tensor case. Finally, we propose to include spatial features as examples of the third modality.

A. AO-ADMM-ASC With Nonnegativity and Sparsity

In CPD, after imposing nonnegativity on the factor matrices, and sparsity and ASC on the abundances, (5) becomes

$$\begin{aligned}&\underset{\boldsymbol{A},\boldsymbol{B},\boldsymbol{C}}{\operatorname{argmin}} \|\boldsymbol{\mathcal{T}} - \boldsymbol{\Lambda} \underset{1}{\bullet} \boldsymbol{A} \underset{2}{\bullet} \boldsymbol{B} \underset{3}{\bullet} \boldsymbol{C}\|_F^2 + \alpha\|\boldsymbol{A}\|_1 \\ &\text{s.t. } \boldsymbol{A} \succeq 0, \quad \boldsymbol{B} \succeq 0, \ \boldsymbol{C} \succeq 0, \ \sum_{r=1}^{R} a_{i,r} = 1 \ |_{\forall i \in \{1,\ldots,I\}}.\end{aligned} \tag{16}$$

In (16), since $\boldsymbol{A}$ adopts the sum-to-one constraint on its rows, it is, hence, enough to normalize only the columns of $\boldsymbol{B}$ and $\boldsymbol{C}$. In principle, these ℓ_2-norms are absorbed in $\boldsymbol{\Lambda}$, but, for the sake of consistency, we use the variable $\boldsymbol{\Psi} = \boldsymbol{\Lambda}\boldsymbol{C}$ instead of $\boldsymbol{\Lambda}$ and $\boldsymbol{C}$, as explained in Section III-C. In order to solve (16), we propose an algorithm inspired by AO-ADMM [48], where the factor matrices are updated in an alternating way and where each update of a factor matrix is optimized as an ADMM subproblem.

1) ASC Solution: In order to model the problem as close as possible to LMM, we need a tensor decomposition algorithm embedding both the ASC and the nonnegativity of factors as constraints. To this end, the strategy that we follow is to extend the concept from NMF to CPD by stacking the following, such that $\sum_{r=1}^{R} a_{i,r} = 1|_{\forall i \in \{1,\ldots,I\}}$ is ensured:

1) a row vector to $\boldsymbol{B}$ (i.e., $\boldsymbol{b}_{J+1,:} \in \mathbb{R}^R$);
2) a lateral slice to $\boldsymbol{\mathcal{T}}$ (i.e., $\boldsymbol{T}_{:,J+1,:} \in \mathbb{R}^{I\times 1\times K}$).

In general, $\boldsymbol{T}_{:,J+1,:}$ can be constructed such that $\forall i \in \{1,\ldots,I\}$ and $\forall k \in \{1,\ldots,K\}$

$$t_{i,J+1,k} = \sum_{r=1}^{R} a_{i,r} b_{J+1,r} \psi_{k,r}. \tag{17}$$

Thus, if we set $\boldsymbol{T}_{:,J+1,K} = \delta \boldsymbol{1}_I$, i.e., $t_{i,J+1,K} = \delta \ \forall i \in \{1,\ldots,I\}$, and $b_{J+1,r} = \delta\psi_{K,r}^{-1} \ \ \forall r \in \{1,\ldots,R\}$, where δ is the mean of $\boldsymbol{\mathcal{T}}$, by substituting the expressions in (17) for $k = K$, we have

$$t_{i,J+1,K} = \sum_{r=1}^{R} a_{i,r} b_{J+1,r} \psi_{K,r} \tag{18a}$$

$$\implies \delta = \delta \sum_{r=1}^{R} a_{i,r}\psi_{K,r}^{-1}\psi_{K,r} = \delta \sum_{r=1}^{R} a_{i,r} \tag{18b}$$

which implies that $\sum_{r=1}^{R} a_{i,r} = 1$.

At the end of each AO-ADMM iteration, $\boldsymbol{\mathcal{T}}$ and $\boldsymbol{B}$ have to be updated, which boils down to a matrix and a vector update after each iteration. We denote by $\tilde{\boldsymbol{\mathcal{T}}}$ and $\tilde{\boldsymbol{B}}$ the extensions of $\boldsymbol{\mathcal{T}}$ and $\boldsymbol{B}$ with the additional lateral slice and row vector, respectively, roughly described as follows:

$$\tilde{\boldsymbol{\mathcal{T}}} = \left[\boldsymbol{\mathcal{T}} \mid \boldsymbol{T}_{:,J+1,:}\right], \ \tilde{\boldsymbol{B}} = \begin{bmatrix}\boldsymbol{B} \\ \boldsymbol{b}_{J+1,:}\end{bmatrix}. \tag{19}$$

Then, expression (16) becomes

$$\begin{aligned}&\underset{\boldsymbol{A},\tilde{\boldsymbol{B}},\boldsymbol{\Psi}}{\operatorname{argmin}} \|\tilde{\boldsymbol{\mathcal{T}}} - \boldsymbol{\mathcal{I}} \underset{1}{\bullet} \boldsymbol{A} \underset{2}{\bullet} \tilde{\boldsymbol{B}} \underset{3}{\bullet} \boldsymbol{\Psi}\|_F^2 + \alpha\|\boldsymbol{A}\|_1 \\ &\text{s.t. } \boldsymbol{A} \succeq 0, \quad \tilde{\boldsymbol{B}} \succeq 0, \ \boldsymbol{\Psi} \succeq 0\end{aligned} \tag{20}$$

where $\boldsymbol{\mathcal{I}}$ is a diagonal tensor of ones.

2) ADMM Updates: At this stage, solving (20) with AO-ADMM becomes simple. We demonstrate the ADMM subproblem updates for each factor matrix starting with $\boldsymbol{A}$.

Supposing that $\tilde{\boldsymbol{T}}_{(1)}$ represents the mode-1 unfolding of $\tilde{\boldsymbol{\mathcal{T}}}$, we can write the subproblem of $\boldsymbol{A}$ as follows:

$$\begin{aligned}\boldsymbol{A} &= \underset{\boldsymbol{A}}{\operatorname{argmin}} \frac{1}{2}\|\tilde{\boldsymbol{T}}_{(1)} - \tilde{\boldsymbol{W}}_{(A)}\boldsymbol{A}^{\mathsf{T}}\|_F^2 + \alpha\|\boldsymbol{A}\|_1 \\ &\text{s.t. } \boldsymbol{A} \succeq 0\end{aligned} \tag{21}$$

where $\tilde{\boldsymbol{W}}_{(A)} \in \mathbb{R}^{(J+1)K\times I} = \tilde{\boldsymbol{B}} \odot \boldsymbol{\Psi}$ represents the Khatri–Rao product [40]. By introducing the splitting variable $\bar{\boldsymbol{A}} = \boldsymbol{A}^{\mathsf{T}}$, expression (21) becomes

$$\begin{aligned}&\underset{\boldsymbol{A},\bar{\boldsymbol{A}}}{\operatorname{argmin}} \frac{1}{2}\|\tilde{\boldsymbol{T}}_{(1)} - \tilde{\boldsymbol{W}}_{(A)}\bar{\boldsymbol{A}}\|_F^2 + \alpha\|\boldsymbol{A}\|_1 \\ &\text{s.t. } \bar{\boldsymbol{A}} = \boldsymbol{A}^{\mathsf{T}} \text{ and } \boldsymbol{A} \succeq 0.\end{aligned} \tag{22}$$

Adopting ADMM for (22), the updates of $\bar{\boldsymbol{A}}$ and $\boldsymbol{A}$ become

$$\begin{aligned}\bar{\boldsymbol{A}} &\leftarrow (\tilde{\boldsymbol{W}}_{(A)}^{\mathsf{T}}\tilde{\boldsymbol{W}}_{(A)} + \rho\boldsymbol{I})^{-1}\left(\tilde{\boldsymbol{W}}_{(A)}^{\mathsf{T}}\tilde{\boldsymbol{T}}_{(1)} + \rho(\boldsymbol{A} + \boldsymbol{U}_{(A)})^{\mathsf{T}}\right) \\ \boldsymbol{A} &\leftarrow \max(0, \bar{\boldsymbol{A}}^{\mathsf{T}} - \boldsymbol{U}_{(A)} - \frac{\alpha}{\rho}) \\ \boldsymbol{U}_{(A)} &\leftarrow \boldsymbol{U}_{(A)} + \boldsymbol{A} - \bar{\boldsymbol{A}}^{\mathsf{T}}\end{aligned} \tag{23}$$

where $\boldsymbol{U}_{(A)} \in \mathbb{R}^{I\times R}$ is called the dual variable.

Similarly, the updates of $\tilde{\boldsymbol{B}}$ and $\boldsymbol{\Psi}$ become

$$\begin{aligned}\bar{\boldsymbol{B}} &\leftarrow \left(\tilde{\boldsymbol{W}}_{(B)}^{\mathsf{T}}\tilde{\boldsymbol{W}}_{(B)} + \rho\boldsymbol{I}\right)^{-1}\left(\tilde{\boldsymbol{W}}_{(B)}^{\mathsf{T}}\tilde{\boldsymbol{T}}_{(2)} + \rho(\boldsymbol{B} + \boldsymbol{U}_{(B)})^{\mathsf{T}}\right) \\ \tilde{\boldsymbol{B}} &\leftarrow \max(0, \bar{\boldsymbol{B}}^{\mathsf{T}} - \boldsymbol{U}_{(B)}) \\ \boldsymbol{U}_{(B)} &\leftarrow \boldsymbol{U}_{(B)} + \tilde{\boldsymbol{B}} - \bar{\boldsymbol{B}}^{\mathsf{T}}\end{aligned} \tag{24}$$

$$\begin{aligned}\bar{\boldsymbol{\Psi}} &\leftarrow (\tilde{\boldsymbol{W}}_{(\Psi)}^{\mathsf{T}}\tilde{\boldsymbol{W}}_{(\Psi)} + \rho\boldsymbol{I})^{-1}\left(\tilde{\boldsymbol{W}}_{(\Psi)}^{\mathsf{T}}\tilde{\boldsymbol{T}}_{(3)} + \rho(\boldsymbol{\Psi} + \boldsymbol{U}_{(\Psi)})^{\mathsf{T}}\right) \\ \boldsymbol{\Psi} &\leftarrow \max\left(0, \bar{\boldsymbol{\Psi}}^{\mathsf{T}} - \boldsymbol{U}_{(\Psi)}\right) \\ \boldsymbol{U}_{(\Psi)} &\leftarrow \boldsymbol{U}_{(\Psi)} + \boldsymbol{\Psi} - \bar{\boldsymbol{\Psi}}^{\mathsf{T}}\end{aligned} \tag{25}$$

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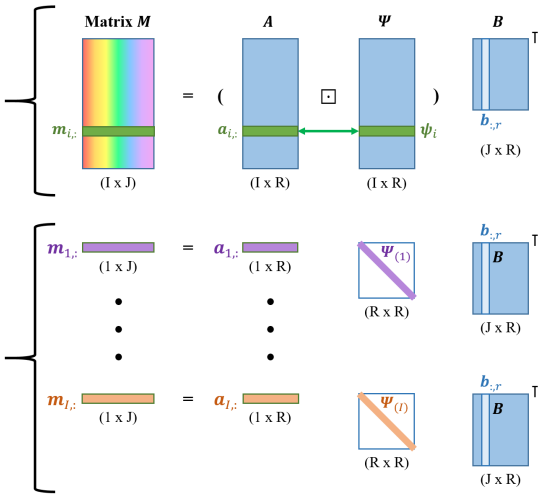


Fig. 4. Our visualization of (12a) (Hadamard product) and (12b) (matrix product). The color code of the bottom part follows that of Fig. 3(b). We have $\boldsymbol{\Psi}_{(i)} = \text{diag}\{\boldsymbol{\Psi}_{i,:}\}$.

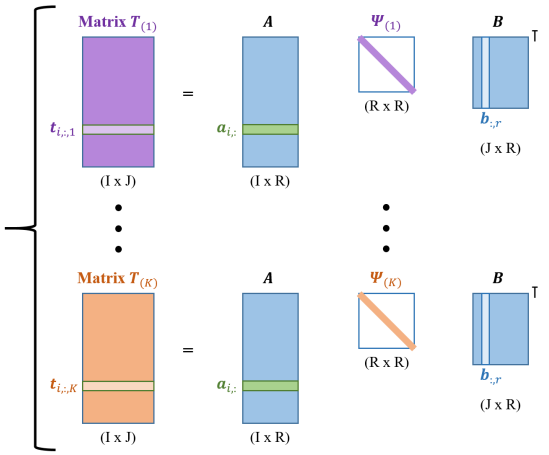


Fig. 5. Our visualization of (14a). The color code is made analogous to that of Fig. 2 and follows that of Fig. 6(a). We have $\boldsymbol{\Psi}_{(k)} = \text{diag}\{\boldsymbol{\psi}_{k,:}\}$.

where $\tilde{\boldsymbol{T}}_{(2)}$ and $\tilde{\boldsymbol{T}}_{(3)}$ are the mode-2 and mode-3 unfoldings of $\tilde{\boldsymbol{\mathcal{T}}}$; $\tilde{\boldsymbol{W}}_{(B)} = \boldsymbol{A} \odot \boldsymbol{\Psi}$ and $\tilde{\boldsymbol{W}}_{(\Psi)} = \boldsymbol{A} \odot \tilde{\boldsymbol{B}}$ are the Khatri–Rao products; and $\boldsymbol{U}_{(B)}$ and $\boldsymbol{U}_{(\Psi)}$ are the dual variables.

Finally, for order-2 tensors, this model becomes equivalent to solving NMF (10). The implementation of AO-ADMM-ASC is summarized in Algorithm 1. The code is available on GitHub in `Python`[4] and `MATLAB`.[5]

Algorithm 1 AO-ADMM-ASC

Require: $\boldsymbol{\mathcal{T}}$, $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{\Psi}$, $\boldsymbol{U}_{(A)}, \boldsymbol{U}_{(B)}, \boldsymbol{U}_{(\Psi)}$, α
  Initialize $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{\Psi}$;
  Initialize $\boldsymbol{U}_{(A)}, \boldsymbol{U}_{(B)}, \boldsymbol{U}_{(\Psi)}$ to zero;
  repeat
    Set $\tilde{\boldsymbol{B}}$ and $\tilde{\boldsymbol{\mathcal{T}}}$ based on (19);
    Set $\boldsymbol{B} \leftarrow \tilde{\boldsymbol{B}}$ and $\boldsymbol{\mathcal{T}} \leftarrow \tilde{\boldsymbol{\mathcal{T}}}$;
    for $\forall \boldsymbol{D} \in \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{\Psi}\}$ **do**
      $\tilde{\boldsymbol{W}}_{(D)} = \odot_{\boldsymbol{J} \neq \boldsymbol{D}} \boldsymbol{J}$;
      $\rho = \text{trace}(\tilde{\boldsymbol{W}}_{(D)}^{\mathsf{T}} \tilde{\boldsymbol{W}}_{(D)})/R$; [48]
      Update $\boldsymbol{D}$ with either (23), (24), or (25);
    end for
    Absorb the columnwise ℓ_2-norms of $\boldsymbol{B}$ into $\boldsymbol{\Psi}$ such that $\boldsymbol{\psi}_{:,r} \leftarrow \boldsymbol{\psi}_{:,r} \|\boldsymbol{b}_{:,r}\|_2 \quad \forall r \in \{1, \ldots, R\}$
    Normalize the columns of $\boldsymbol{B}$
  until Termination criterion (e.g., number of iterations)
  return $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{\Psi}$

B. Interpreting Tensor-Based Unmixing, ASC, and ELMM

Here, we build upon what has been presented in Sections III-B and III-C as methodological, physical, and graphical bases for the MultiHU-TD interpretation. For that, we first draw the analogies between the expressions of Sections III and IV by starting from the interpretation of the matrix case and elaborating that of the tensor case. In the process, we break down the physical meaning of ASC and that of the so-called SV function of ELMM in MultiHU-TD at the base of the composition of the frontal slices of the tensor. Then, we visualize the expressions in order to interpret MultiHU-TD through graphical representations of subspaces while commenting on the physical role of the extracted factors and the number of latent components R.

1) Interpretation of ASC: In the matrix case, assuming a matricized HSI $\boldsymbol{M}$ (that is, after reordering the two pixel modes into one mode in lexicographic order) such that

$$\boldsymbol{M} = \boldsymbol{A}\,\boldsymbol{B}^{\mathsf{T}} \tag{26}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ represent the estimated abundances and EMs, respectively, the physical meaning of ASC is that it constrains the columns of $\boldsymbol{B}$ to form a simplex. Then, the rows of $\boldsymbol{A}$ (which sum to one) represent the position of the pixels on said simplex. This is visualized in Figs. 3 and 4.

In the tensor case, we assume a tensor $\boldsymbol{\mathcal{T}}$ whose CPD is expressed as

$$\boldsymbol{\mathcal{T}} = \boldsymbol{\mathcal{I}} \underset{1}{\bullet} \boldsymbol{A} \underset{2}{\bullet} \boldsymbol{B} \underset{3}{\bullet} \boldsymbol{\Psi} \tag{27}$$

where $\boldsymbol{\mathcal{I}}$ is a diagonal tensor of ones, $\boldsymbol{A}$ and $\boldsymbol{B}$ represent the estimated abundances and EMs, respectively, and $\boldsymbol{\Psi}$ represents the factor matrix of the third modality.

Here, each slice of the tensor $\boldsymbol{\mathcal{T}}_{:,:,k}$ $\forall k \in \{1, \ldots, K\}$ represents a matricized HSI similar to $\boldsymbol{M}$ (e.g., corresponding to acquisitions at different dates in a time series, at different angles in multiangular acquisitions, or at different scales in a multiscale decomposition), and we have the following linear relationship:

$$\boldsymbol{\mathcal{T}}_{:,:,k} = \boldsymbol{A}\,(\text{diag}\{\Psi_{k,:}\}\,\boldsymbol{B}^{\mathsf{T}}) = \boldsymbol{A}\,\boldsymbol{B}^{(k)\mathsf{T}} \quad \forall k \tag{28}$$

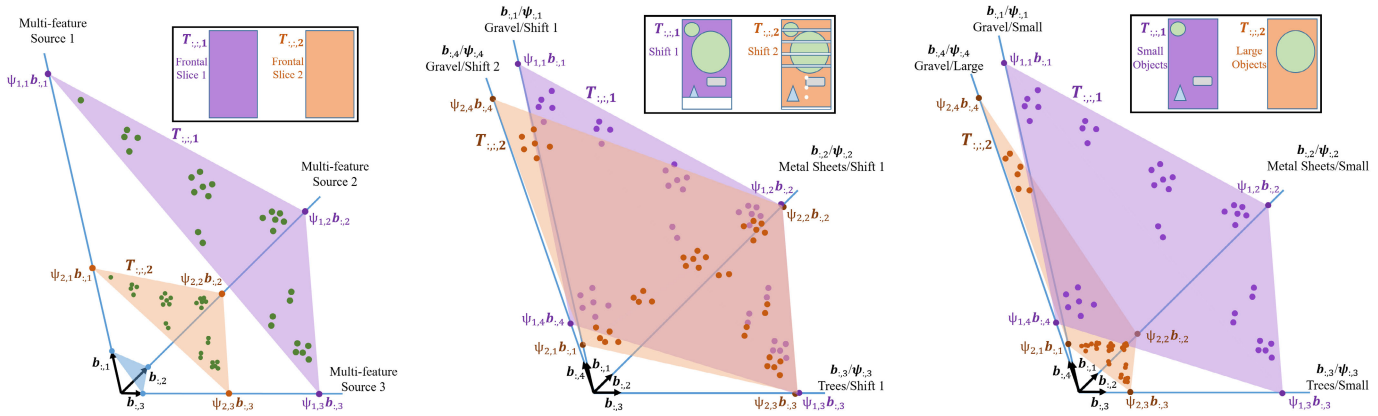
where $\Psi_{k,:}$ represents the kth row of $\boldsymbol{\Psi}$. First, let us look at the properties of the aforementioned expression.

1) If we look at each slice separately, the physical meaning of applying ASC is similar to that of the matrix case where the simplex is formed out of the columns of $\boldsymbol{B}^{(k)}$. In fact, we have that $\boldsymbol{B}^{(k)} = \boldsymbol{B}\ \text{diag}\{\Psi_{k,:}\}$, which means that the columns of $\boldsymbol{B}^{(k)}$ are only scaled versions of those of $\boldsymbol{B}$ such that

$$\boldsymbol{b}^{(k)}_{:,r} = \psi_{k,r}\boldsymbol{b}_{:,r} \quad \forall r \in \{1, \ldots, R\}. \tag{29}$$

[4]https://github.com/mhmdjouni/MultiHU-TD-Python
[5]https://github.com/mhmdjouni/MultiHU-TD-MATLAB

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(a) ELMM-CPD (General) (b) ELMM-CPD (Patches) (c) ELMM-CPD (MM)

Fig. 6. Graphical representations of (a) CPD with $R = 3$ components in the case of three spectral signatures $\{\boldsymbol{b}_{:,1}, \boldsymbol{b}_{:,2}, \boldsymbol{b}_{:,3}\}$ and two frontal slices $\{\boldsymbol{T}_{:,:,1}, \boldsymbol{T}_{:,:,K}\}$, (b) Patch-CPD, and (c) MM-CPD with $R = 4$ components in the case of four spectral signatures $\{(\boldsymbol{b}_{:,1}, \boldsymbol{\psi}_{:,1}), \ldots, (\boldsymbol{b}_{:,4}, \boldsymbol{\psi}_{:,4})\}$ and two frontal slices $\{\boldsymbol{T}_{:,:,1}, \boldsymbol{T}_{:,:,2}\}$. Very correlated vectors can be seen as "spectral bundles" with different third-modality characteristics. The relative coordinates of the pixels in the convex hulls must be the same since $\boldsymbol{A}$ is common for all the frontal slices. (a) ELMM-CPD (general). (b) ELMM-CPD (patches). (c) ELMM-CPD (MM).

2) If we consider all the slices together, we notice that the abundance matrix $\boldsymbol{A}$ is common to all of them. Moreover, we notice that the factor matrix of estimated EMs $\boldsymbol{B}$ (obtained through CPD) is at the base of their estimated EMs, influenced only by the corresponding scaling factors in $\Psi_{k,:}$, which encode the corresponding third-mode features.
3) Given that $\boldsymbol{A}$ and $\boldsymbol{B}$ factorize the physical and data structures along the pixel and spectral mode, respectively, they are independent of the third-mode differences in the hyperspectral scene between the slices. One could even construct a matrix $\boldsymbol{M}^{(\text{CPD})}$ from the first two factor matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ (obtained through CPD) such that

$$\boldsymbol{M}^{(\text{CPD})} = \boldsymbol{A}\,\boldsymbol{B}^{\mathsf{T}} \tag{30}$$

where $\boldsymbol{B}$ is independent of the spectral variabilities present along the third mode.
4) As $\boldsymbol{A}$ and $\boldsymbol{B}$ factorize the pixel and spectral information, the third-mode factor matrix $\boldsymbol{\Psi}$ encodes the changes between the slices along the third mode where the kth row $\Psi_{k,:}$ is associated with the kth slice. Intuitively speaking, this allows some degrees of freedom to express the nonlinearities along the third mode in a linear sense, which is reflected on the level of each slice by scaling the columns of $\boldsymbol{B}$, as expressed in (29).

Finally, assuming that we have K slices in the tensor, imposing ASC is the equivalent of having K simplices whose edges, defined by the columns of $\boldsymbol{B}^{(k)}$ $\forall k \in \{1, \ldots, K\}$, can move only along the directions of the columns of $\boldsymbol{B}$ such that the following holds.

1) The positions of the edges of each simplex are defined by the corresponding scaling factor $\psi_{k,r}$, which encodes the third-mode physical property (whether it is time, morphological properties such as scale, brightness, and neighborhood pixels) of the rth estimated EM $\boldsymbol{b}_{:,r}$ in the kth tensor slice $\boldsymbol{T}_{:,:,k}$.
2) For instance, an estimated EM $\boldsymbol{b}_{:,r}$ can be relevant in a given slice $\boldsymbol{T}_{:,:,1}$ due to a high factor $\psi_{1,r}$, such that $\boldsymbol{b}^{(1)}_{:,r} = \psi_{1,r}\boldsymbol{b}_{:,r}$, but also have a low contribution in another slice $\boldsymbol{T}_{:,:,2}$ due to a low factor $\psi_{2,r}$, such that $\boldsymbol{b}^{(2)}_{:,r} = \psi_{2,r}\boldsymbol{b}_{:,r}$. We can see that $\boldsymbol{b}_{:,r}$ remains independent of the physical entity that the third mode represents but also that $\boldsymbol{b}^{(1)}_{:,r}$ and $\boldsymbol{b}^{(2)}_{:,r}$ move along the direction of $\boldsymbol{b}_{:,r}$ (due to the scaling factors $\psi_{1,r}$ and $\psi_{2,r}$) based on the effect that the third-mode physical entity applies on $\boldsymbol{b}_{:,r}$ in the given slice.
3) Since $\boldsymbol{A}$ is common to all the slices, the positions of the pixels are relatively fixed to each of the K simplices.

This is demonstrated in Figs. 5 and 6.

2) Interpretation of Tensor-Based ELMM: First, we note that expressions (14a) and (14b) are analogous to the ELMM expression (12b). The major difference between the two cases is that, in CPD, the scaling factors are *frontal slice-dependent* ($\psi_{k,r}$), while, in ELMM, they are pixel-dependent ($\psi_{i,r}$). Second, we visualize (12a) and (12b) in Fig. 4 and (14a) and (14b) in Fig. 5. Looking at (14a), the frontal slices $\boldsymbol{T}_{:,:,k}$ and the physical meaning that they represent have a direct influence on the SV function $\boldsymbol{f}_k$ and the interpretation of the SVs, which is simply reflected as scaling factors in each row of $\boldsymbol{\Psi}$, i.e., $\boldsymbol{\psi}_{k,:}$ (or $\boldsymbol{\Psi}_{(k)} = \text{diag}\{\boldsymbol{\psi}_{k,:}\}$).

As a result, since the spatial information and the spectral information are factorized and represented by $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively, and since $\boldsymbol{A}$ and $\boldsymbol{B}$ are shared by all the frontal slices, then each frontal slice $\boldsymbol{T}_{:,:,k}$ is inherently differentiated through a set of R scaling factors $\{\psi_{k,1}, \ldots, \psi_{k,R}\}$. Consequently, the spectral information in each frontal slice $\boldsymbol{T}_{:,:,k}$ can be seen as the set of scaled sources $\{\boldsymbol{b}^{(k)}_{:,r} = \psi_{k,r}\boldsymbol{b}_{:,r}\}|_{\forall\, r\in\{1,\ldots,R\}}$, where $\{\boldsymbol{b}_{:,r}\}$ are the columns of $\boldsymbol{B}$ (independent of the slices), $\boldsymbol{b}^{(k)}_{:,r}$ are their spectral variations per frontal slice, and $\boldsymbol{\psi}_{k,:}$ encodes the scaling factors of these variations.

This also means that R is a major parameter that represents the degrees of freedom, especially through the scaling factors of $\boldsymbol{\Psi}$, which then jointly encodes the following:

1) the mode-3 evolution of the extracted components of $\boldsymbol{A}$ and $\boldsymbol{B}$ in its columns $\boldsymbol{\psi}_{:,r}$;
2) the per-slice modeling of the SVs in its rows $\boldsymbol{\psi}_{k,:}$.

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Intuitively, when obtaining an augmented HSI tensor, one can say that the physical representations of any applied transformations (e.g., scale and illumination) [30] and any natural evolution of a scene (e.g., time series) [24], and resulting in $\boldsymbol{\mathcal{T}}_{:,:,k}$, are reflected and observed through the matrix $\boldsymbol{\Psi}$ of the decomposition. Moreover, we point out the following.

1) In CPD, there are as many SV functions (and simplices) as the frontal slices of the tensor, which is significantly lower than the number of pixels ($K \lll I$).
2) In CPD, one row of R scaling factors in $\boldsymbol{\Psi}$ corresponds to a full frontal slice and is shared by all the pixel rows of $\boldsymbol{A}$, while, in classical ELMM, each row of scaling factors in $\boldsymbol{\Psi}$ corresponds to one pixel of $\boldsymbol{M}$ and interacts with only one row of $\boldsymbol{A}$. This clearly appears when we compare (11) and (12b) with (14a) and (14b), and Fig. 4 with Fig. 5.
3) On a graphical representation, when ASC is imposed, CPD suggests that each simplex contains I pixels, as illustrated in Fig. 6(a), such that the relative coordinates of the pixels inside each convex hull are the same since each row of $\boldsymbol{\Psi}$ interacts with all of $\boldsymbol{A}$.

In summary, having a third mode in HSI produces scaling factors in ELMM that absorbs the SVs based on the physical meaning of the frontal slices along the third mode (e.g., time, patches, and MM), which balances the extracted factors in $\boldsymbol{A}$ and $\boldsymbol{B}$ independently of said SVs. Moreover, the imposed value of R represents the number of extracted sources and scaling factors and has a major effect on the results and the SV interpretation. Since the extraction of materials is also driven by the third mode diversity, it is possible to expect a few more latent components than the number of pure "EMs" existing in the scene, with some spectral correlations, as roughly demonstrated in Fig. 6. As R decreases, we tend toward having fewer degrees of freedom, where CPD tends toward extracting the EMs while applying a regularization on the observed pixels influenced by the physical meaning of the information across the third mode. As R increases, we tend toward having more degrees of freedom, where CPD tends toward extracting factors with multifeature separability of the sources. We note that R should not be too high in order to avoid overfitting[6] (given the relative low-rankness of the data) and to ensure the uniqueness of the CPD [40], [50].

C. Examples of Third-Mode Features: Spatial Features (Patches/MM)

In this section, for the sake of comparison and illustration, we consider two examples of spatial features that augment an HSI into a third-order tensor for MultiHU-TD: neighborhood patches and MM. Having these two types of features allows for a more comprehensive comparison in terms of the properties of the MultiHU-TD framework and its links to previous works on said features. We revisit the case of patches with additional insights and introduce MM. Consequently, this helps demonstrate the interpretability of the model, especially in terms of physical significance and the variation of R. We often refer to Figs. 5 and 6 for illustration.

1) Patches: We recall that the motivation for adding patches as features is to perform a spatial regularization by considering the spatial correlation of neighboring pixels [14]. Constructing an HSI tensor from neighborhood patches (coined as "Patch-HSI tensor") is illustrated in Fig. 7 for a 5×5 patch. In short, each pixel tube in the original HSI cube is taken with a patch of its neighboring pixels (of predetermined size), and then, the pixel and its neighbors are stacked as a horizontal slice $\boldsymbol{\mathcal{T}}_{i,:,:}$ in the third-order tensor. That said, the first frontal slice $\boldsymbol{\mathcal{T}}_{:,:,1}$ of said tensor is usually the matricized HSI (where the pixels represent one mode).

Here, we add that a Patch-HSI tensor has an inherently low-rank structure, which is equal to that of the NMF of the matricized HSI, that is, the information contained along the third mode by each of the frontal slices of the tensor is almost essentially the same. In fact, the frontal slices are just *spatially shifted versions* of the original image $\boldsymbol{\mathcal{T}}_{:,:,1}$, and the values of these shifts correspond to a *small spatial kernel*, usually around 3×3 or 5×5. However, what is different in the tensor case is that, when this *shifting* information is stacked along the third mode and CPD is imposed with such a low value of R, the model automatically applies an implicit smoothing of the pixels that belong to the same patch (i.e., the same horizontal slice of $\boldsymbol{\mathcal{T}}$). This is because the frontal slices are jointly factorized with the degrees of freedom of a single one of them while also sharing the information of $\boldsymbol{A}$ and $\boldsymbol{B}$.

Therefore, a main advantage over NMF is that one expects to extract the same sources with a patch-local smoothing of the SVs of the estimated EMs, where the SVs are balanced out in the form of scaling factors stored in the rows of $\boldsymbol{\Psi}$. An important note here is that the scaling factors stored in $\boldsymbol{\Psi}$ may not have a significant physical meaning.

Now, what happens when R increases? Since the information across the frontal slices is essentially the same (implying redundancy), the sources and abundances are expected to replicate, and we expect to observe slightly spatially shifted versions of the AMs (i.e., in the columns of $\boldsymbol{A}$). In this case, the scaling factors in $\boldsymbol{\Psi}$ only indicate whether an estimated EM in $\boldsymbol{B}$ corresponds to a certain spatial shifting or another. This point is roughly illustrated in Fig. 6(b) (inspired by Fig. 11) where we have three spectral sources: Gravel, Metal Sheets, and Trees, but CPD is carried out with $R = 4$. Here, the convex hull of $\boldsymbol{\mathcal{T}}_{:,:,1}$ gives a high scaling factor at $\boldsymbol{b}_1$ and a low factor at $\boldsymbol{b}_4$, while that of $\boldsymbol{\mathcal{T}}_{:,:,2}$ gives the opposite with almost the same quantity. This is due to the fact that the materials are present in almost the same quantity in both frontal slices. In other words, there may be a problem of redundancy if some components account for the same material with patches, which does happen in practice.

This problem does not occur when the third mode represents a physical meaning such as in the case of MM.

2) Mathematical Morphology: While using patches is efficient, it still ignores the physical properties of connected pixels, and the SVs are regularized indifferently among pixels

[6]There is no exact value of the tensor rank, and finding a good estimate is an open challenge (as for matrix factorization problems), which is out of the scope of this article. As a rule of thumb, R could be chosen by looking at the reconstruction error in the factorization, e.g., by choosing the smallest value of R providing an acceptable reconstruction, or by observing the elbow of the plot of singular values of the mode-1 unfolding of the data.

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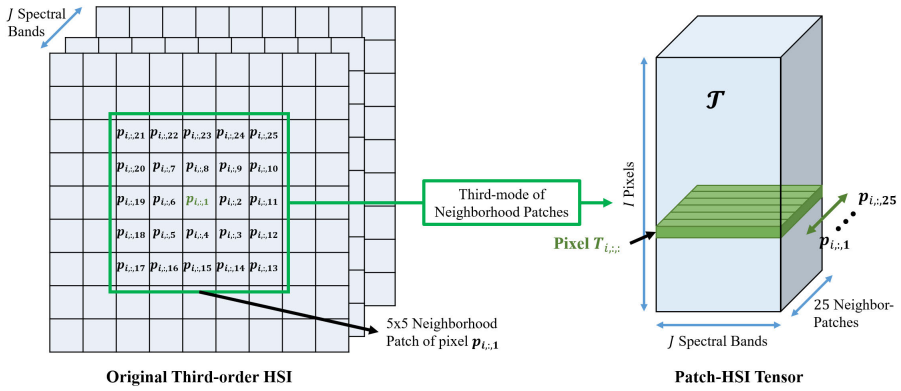


Fig. 7. Illustration of constructing a 5 × 5 Patch-HSI tensors based on [14].

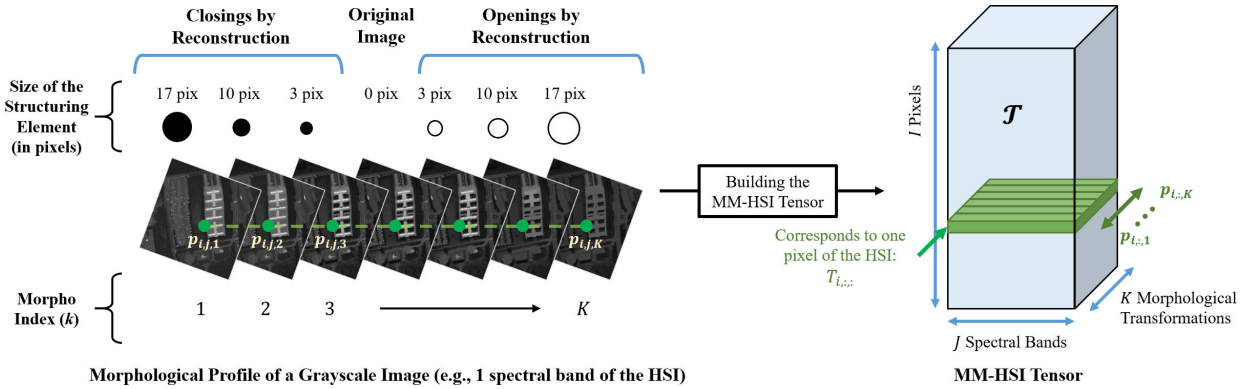


Fig. 8. Example of sequential morphological filtering of a gray-scale image (corresponding to one spectral band of the HSI of Pavia University) with openings and closings by reconstruction using successive sizes of the SE, which is a disk in this case. The stacking of the transformations, with the original image corresponding to 0 size being placed in the middle, is referred to as the morphological profile of the image. The green spots mark the same pixel positioning in each of the transformations. Then, to create the MM-HSI tensor [30], the pixel positioning modes are reordered into lexicographic order along the first mode, and the yellow spots are stored along the third mode in order to create the tensor.

belonging to different types of materials. On the other hand, morphological features [56], [57] take into account physical properties such as scale and brightness of objects and promote dealing with SV among pixels sharing these properties. Constructing an HSI tensor using MM (coined as "MM-HSI tensor") is illustrated in Fig. 8. In short, the matricized versions of the original HSI and the results of its morphological transformations are stacked as the frontal slices of the tensor.[7]

Through MM, we emphasize the role of incorporating spatial diversities that add physical significance to the objects of the scene. As such, one expects that an MM-HSI tensor has a more complicated structure than that of a Patch-HSI tensor since its frontal slices contain additional context on the materials, such as their sizes and brightness levels. As such, imposing a low R promotes spectral smoothing of the SVs based on a morphological regularization of the abundances, while imposing a sufficiently high R promotes a distinctive spectral–morphological multifeature separation of the materials, unlike Patch-HSI tensors.

Since each frontal slice is seen as a characteristic of spatial scale (i.e., size of objects in the scene) and/or brightness (which is particularly relevant for ELMM because scaling factors can be directly linked to brightness), then the scaling factors represented by $\boldsymbol{\Psi}$ indicate the quantitative correspondence of an extracted material to the aforementioned physical properties per frontal slice. This point is roughly illustrated in Fig. 6(c) (inspired by Fig. 12) where $\boldsymbol{T}_{:,:,1}$ and $\boldsymbol{T}_{:,:,2}$ characterize small and large objects, respectively. Here, the convex hull of $\boldsymbol{T}_{:,:,1}$ gives high scaling factors for $\boldsymbol{b}_1$, $\boldsymbol{b}_2$, and $\boldsymbol{b}_3$ (corresponding to small objects), and a low scaling factor at $\boldsymbol{b}_4$ (corresponds to large objects), while that of $\boldsymbol{T}_{:,:,2}$ gives the opposite.

[7]The details of constructing MM-HSI tensors are out of the scope of this article but can be found in [30, Sec. III].

V. Experiments and Results

In this section, we discuss the experiments and results of MultiHU-TD on real HSIs in terms of AO-ADMM-ASC (compared to Naive ASC [14]), extracted factors, ELMM interpretability, and qualitative comparisons between Patch-HSI and MM-HSI tensors with low and high values of the number of latent components. In each experiment, among 30 random initializations of the factor matrices, the result with the minimum RMSE is chosen based on (31a). The estimated EMs of $\boldsymbol{B}$ are identified based on their minimum SAD, in degrees, with respect to the reference EMs based on (31b)

$$\text{RMSE}(\mathcal{T}, \hat{\mathcal{T}}) = \frac{\|\mathcal{T} - \hat{\mathcal{T}}\|_F^2}{\|\mathcal{T}\|_F^2} \tag{31a}$$

$$\text{SAD}(\boldsymbol{e}, \boldsymbol{b}) = \arccos\left(\frac{\boldsymbol{e} \cdot \boldsymbol{b}}{\|\boldsymbol{e}\|_2 \cdot \|\boldsymbol{b}\|_2}\right). \tag{31b}$$

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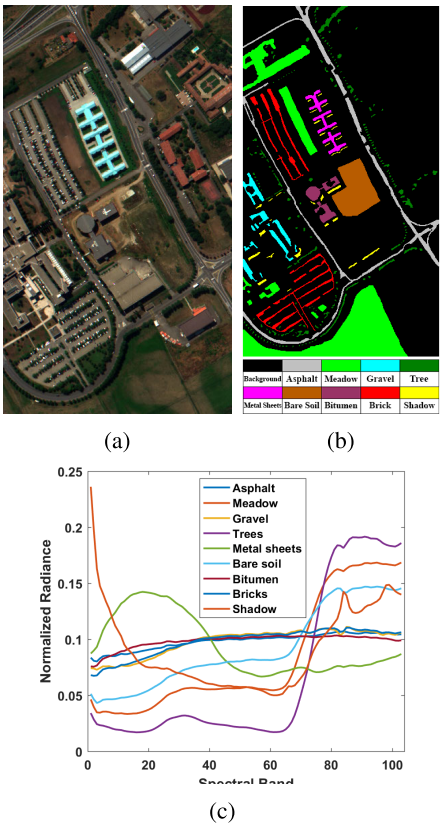


Fig. 9. (a) Pavia in false colors. (b) Pavia's spatial reference. (c) Pavia's spectral reference extracted by averaging each class of the spatial reference.

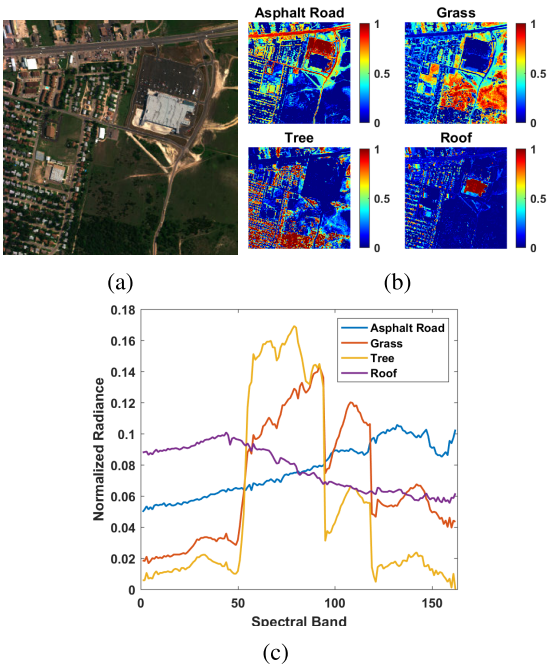


Fig. 10. (a) Urban in false colors. (b) Urban's spatial reference. (c) Urban's spectral reference.

We want to consider HSIs that show objects with spatial features of different scales and brightness levels, for which urban areas are good candidates. For that, we choose two real HSIs[8]: *Pavia University* and *Urban*, as shown in Figs. 9 and 10, respectively, in false colors with their spatial and spectral references. It is worth noting for the sake of interpretation that the MM-HSI tensors are built following the EMP technique used in [30] with ObRs and CbRs, which correspond to bright and dark objects, respectively, with varying sizes of the SE, which, in turn, corresponds to the scales of objects. In the following experiments, four sizes of the SE are used, corresponding to eight ObR and CbR transformations, and then, the dimension of the third mode is $K = 9$ where the original image (corresponding to scale 0) is placed in the middle, as roughly shown in Fig. 8. The Patch-HSI tensors are built following [14], as shown in Fig. 7, with 3×3 patches. This means that the dimension of the third mode is $K = 3 \times 3 = 9$, where the original image is placed at $k = 1$.

The maps and plots shown in the experiments represent the *columns* of the factor matrices. Above each AM, we show the material that corresponds to it with its minimum SAD value. We recall that a set of similarly indexed columns, e.g., $\{\boldsymbol{A}_1, \boldsymbol{B}_1, \text{and } \boldsymbol{\Psi}_1\}$, represent the abundance, spectral source signal, and third-mode source pattern (e.g., morphological print and shifting print) of one extracted material, respectively. Since $\boldsymbol{\Psi}$ plays a crucial role in the interpretability of ELMM and MultiHU-TD, we also highlight the relevance of its row components. Each fixed index k in the plot corresponds to a row of $\boldsymbol{\Psi}$ and, thus, to a frontal slice in $\boldsymbol{\mathcal{T}}$ and the vertical grouping of points at said index, as indicated in Figs. 11(c), 12(c), 13(c), and 14(c), represents the scalars in that row. This also means that, when projected on Fig. 6, in Patch-CPD, $k = 1$ contains the SV scaling factors that balance the convex hull of the original HSI and are responsible for its reconstruction, while, in MM-CPD, the middle index does that. This will eventually show how MM accounts for physical effects in the scene, while patches do not.

That said, we note that we carried part of the experiments using sparse NMF with ASC [17] for the sake of qualitative comparison of the AMs and spectral sources obtained from the original HSI (i.e., the HSI matrix without additional filtering). Due to the difference in the type of information contained between the matrix and tensor cases, and since NMF does not apply in the framework of MultiHU-TD, these results will serve only as a reference for the extracted components of CPD as they do not serve the main aim and message of this work. For that reason, we include them in Appendix A with further reasoning and explanation of the NMF case analysis and its relevance to this work.

In each case analysis, we look into the components of $\boldsymbol{A}$ and $\boldsymbol{B}$ first, which visually and spectrally identify the materials, and then, we explain their correspondence to those of $\boldsymbol{\Psi}$, where we are interested in the significance of the third-mode patterns and then their relevance to the original HSI. We note that quantitative validation of the AMs and EMs is usually not evident, especially in the case of the Urban HSI in Fig. 11(b), where the spatial GT is not a real GT but actually just a reference and is not usable for quantitative comparison. Moreover, there is neither a quantitative nor a

[8]The datasets with detailed information are available on the website: http://lesun.weebly.com/hyperspectral-dataset.html.

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TABLE V
PAVIA. THE RESULTS OF AO-ADMM-ASC AND NAIVE ASC IN TERMS OF RMSE AND EXECUTION TIME: R INDICATES THE NUMBER OF LATENT COMPONENTS. THE RESULTS OF THE MINIMUM RMSE ARE SHOWN

Algorithm	R	RMSE %	Time (s)
Naive ASC [14]	8	7.07	231
AO-ADMM-ASC	8	6.34	384

qualitative reference for third-mode patterns in the literature, highlighting an in-depth qualitative analysis.

In addition to the real HSIs, we demonstrate the performance of the proposed framework under different levels of noise through a synthetic HSI tensor in Appendix B. We run our experiments with Intel[9] Core[10] i7-1185G7, 32-GB RAM, 3200-MHz LPDDR4.

A. Results Discussion—Pavia University

In this section, we present the experiments of the HSI of Pavia, but, first, we note that the reference for this dataset is originally composed of training and testing sets, where each pixel is manually associated with one of nine labeled classes, as seen in Fig. 9(b). The spectral reference in Fig. 9(c) is extracted by averaging the spectral signatures of each subset of pixels belonging to one class. In Fig. 9(c), some classes have very similar spectral signatures, so, in the following, sometimes, we refer to *Trees* and *Meadows* as *vegetation* and to *Asphalt*, *Bitumen*, *Gravel*, and *Bricks* as *roads* or *roofs*, while *Bare Soil* may belong to either of both groups.

We start by comparing AO-ADMM-ASC and Naive ASC. After that, we focus on CPD and the ELMM analysis of the factors while interpreting the cases of patches and MM. For the MM-HSI tensor, our SEs are disks with the successive radii: $\{2, 7, 12, 17\}$ pixels. Both Patch- and MM-HSI tensors then have $K = 9$ frontal slices and dimensions $207400 \times 103 \times 9$. Finally, we find that $R = 4$ and $R = 8$ are the best for low and high values of the number of latent components, respectively.

1) AO-ADMM-ASC: Here, we compare the RMSE results of MM-CPD between AO-ADMM-ASC and Naive ASC [14]. The results are shown in Table V, where we see that, with AO-ADMM-ASC, we gain in RMSE, which corresponds to a better estimation of the factors with respect to the observed tensor with a small difference in the execution time.

2) Few Latent Components, ELMM, and SV: Here, we are interested in the property of CPD dealing with SV. Since Patch-HSI tensors have an inherently low structure as that of NMF, we start by considering $R = 4$ for patches and MM. As explained in Section IV-B, such a low value highlights the SV and *spatial regularization* aspects of MultiHU-TD. We compile the discussion into two stages: 1) Fig. 11 represents Patch-CPD and 2) Fig. 12 represents MM-CPD.

[9]Registered trademark.
[10]Trademarked.

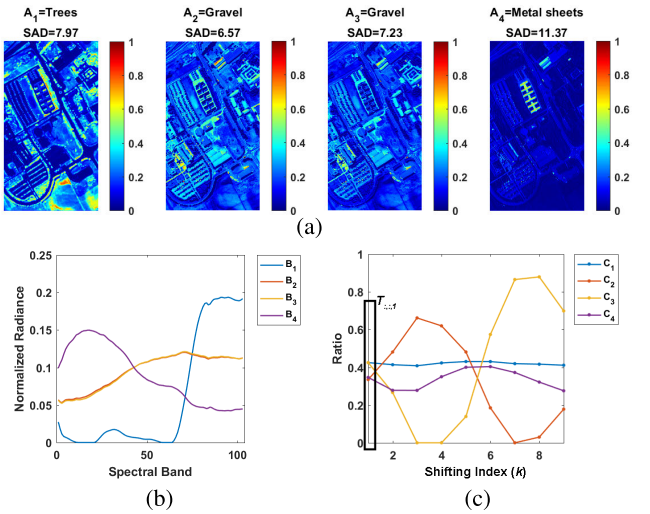


Fig. 11. Pavia. CPD results of the patch-tensor for $R = 4$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$. (c) Components of $\boldsymbol{\Psi}$.

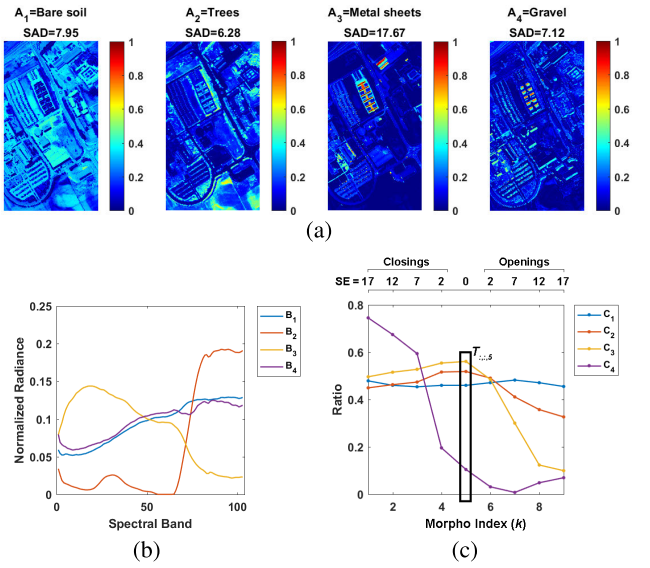


Fig. 12. Pavia. CPD results of the MM-tensor for $R = 4$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$. (c) Components of $\boldsymbol{\Psi}$.

First, in Fig. 11, $\boldsymbol{B}_1$ and $\boldsymbol{B}_4$ are identified as *Trees* and *Metal Sheets*, respectively, which reflects the areas highlighted in $\boldsymbol{A}_1$ and $\boldsymbol{A}_4$, while $\boldsymbol{B}_2$ and $\boldsymbol{B}_3$ can be seen as a bundle identified as *Gravel*, which reflects the areas highlighted in $\boldsymbol{A}_2$ and $\boldsymbol{A}_3$ and where we already start to see *replicated* components. In fact, $\boldsymbol{A}_2$ and $\boldsymbol{A}_3$ are *slightly shifted* versions of each other, which is explained better in the following interpretation of $\boldsymbol{\Psi}_2$ and $\boldsymbol{\Psi}_3$.

Looking at Fig. 11(c), which is the most interesting, one might intuitively expect to see horizontal curves since, quantitatively, the collective SV is supposedly *constant* in patches. However, while $\boldsymbol{\Psi}_1$ and $\boldsymbol{\Psi}_4$ look almost straight, $\boldsymbol{\Psi}_2$ and $\boldsymbol{\Psi}_3$ are not. We notice here that, where $\boldsymbol{\Psi}_2$ is high, $\boldsymbol{\Psi}_3$ is low and vice versa. In part, this means that $\boldsymbol{A}_2$ represents the shifts where $k = \{2, 3, 4, 5\}$, while $\boldsymbol{A}_3$ represents those where $k = \{6, 7, 8, 9\}$. In another part, the two columns fluctuate in a way that *maintains* a constant SV and *balances out* their

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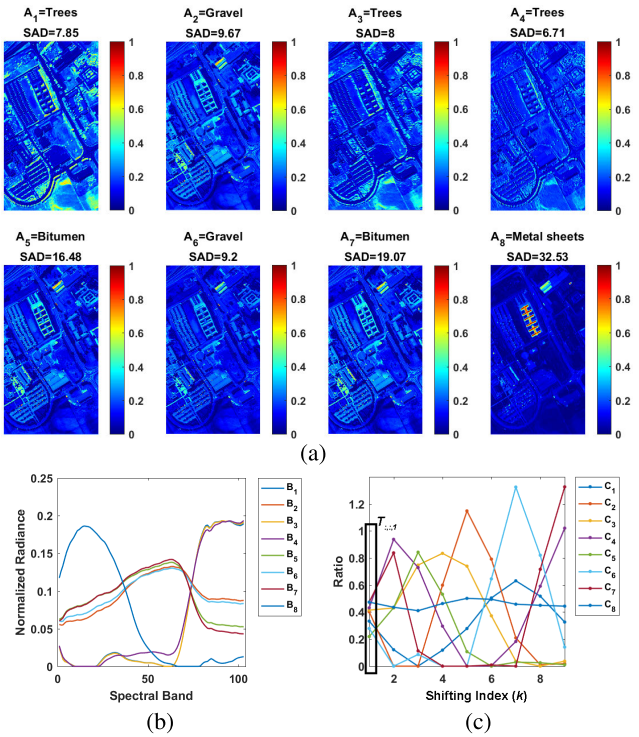


Fig. 13. Pavia. CPD results of the patch-tensor for $R = 8$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$. (c) Components of $\boldsymbol{\Psi}$.

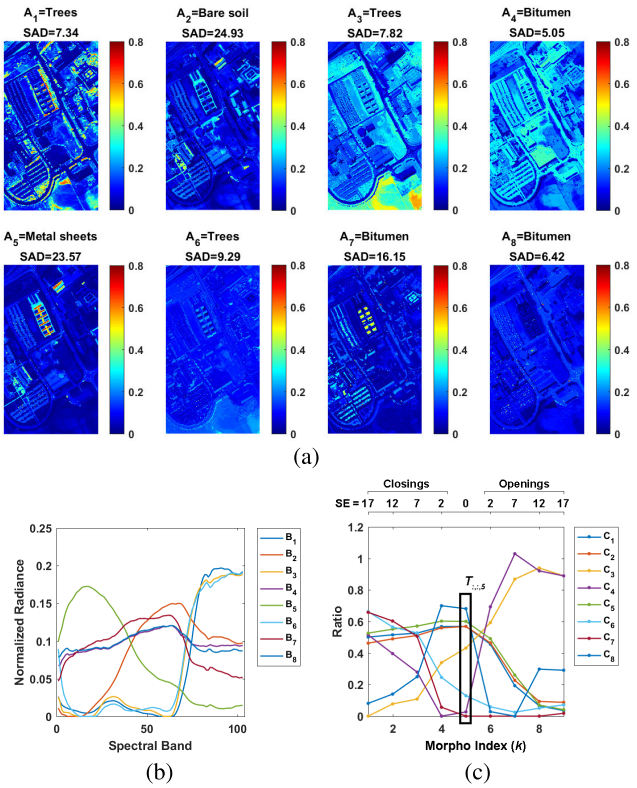


Fig. 14. Pavia. CPD results of the MM-tensor for $R = 8$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$. (c) Components of $\boldsymbol{\Psi}$.

quantitative presence across the frontal slices. As for $k = 1$, which represents the original HSI, we notice that the scaling factors are almost equal, which means that the spectral vectors of the convex hull are *equally present* in the HSI, all of which show that Patch-HSI tensors do not account to physical spatial effects.

Second, in Fig. 12, $\boldsymbol{B}_1$, $\boldsymbol{B}_2$, and $\boldsymbol{B}_3$ are identified as *Bare Soil*, *Trees*, and *Metal Sheets*, respectively, which reflects the areas highlighted in $\boldsymbol{A}_1$, $\boldsymbol{A}_2$, and $\boldsymbol{A}_3$, all of which is similar to those obtained by patches. As for $\{\boldsymbol{A}_4, \boldsymbol{B}_4\}$, while $\boldsymbol{B}_4$ and $\boldsymbol{B}_1$ can be seen as a spectral bundle, unlike patches, we notice that $\boldsymbol{A}_4$ highlights interesting *shadow* areas (i.e., dark features), which clearly reflects the *morphological awareness* incorporated into CPD with MM. The latter becomes more interesting with the following interpretation of $\boldsymbol{\Psi}$.

Looking at Fig. 12(c), we observe three main patterns that can be associated with the chosen morphological parameters. First, $\boldsymbol{\Psi}_4$ corresponds to dark features (reflected by the shadows in $\boldsymbol{A}_4$) as it has higher values when k corresponds to CbR and then continues decreasing toward ObR. Second, $\boldsymbol{\Psi}_2$ and $\boldsymbol{\Psi}_3$ correspond to small features as they have higher values around the middle ($k = 5$) where the SEs are small, which is visually reflected through the small objects highlighted in $\boldsymbol{A}_2$ (trees) and $\boldsymbol{A}_3$ (metal sheets and vehicles). Third, $\boldsymbol{\Psi}_1$ is rather steady, which means that the spatial features shown in $\boldsymbol{A}_1$ are general.

As for $k = 5$, which represents the original HSI, we notice that $\boldsymbol{\Psi}_2$ and $\boldsymbol{\Psi}_3$ have the highest scaling factors since they correspond to relatively bright objects of the scene, $\boldsymbol{\Psi}_1$ has a slightly lower factor since it corresponds to darker objects, such as asphalt roads, building roofs, parking lots, and bare soil areas, and $\boldsymbol{\Psi}_4$ has the lowest factors since it corresponds to dark shadows. These relationships show the columnwise and rowwise significance of $\boldsymbol{\Psi}$ and how MultiHU-TD can balance out the SVs and simultaneously reconstruct the original HSI.

3) More Latent Components, ELMM, and Multifeature Separability: Here, we are interested in MultiHU-TD when we have more degrees of freedom, where we dive deeper into the factors of patches and MM for $R = 8$. As explained in Section IV-B, such a higher value demonstrates the multifeature separability of MM and how patches only replicate its components. We compile the discussion into two stages: 1) Fig. 13 represents Patch-CPD and 2) Fig. 14 represents MM-CPD.

First, in Fig. 13, we end up with more replicas of the same $\boldsymbol{A}$ and $\boldsymbol{B}$ components obtained in Fig. 11, the bundles being associated with the column indices $\{1, 3, 4\}$ detected as *Trees* and $\{2, 5, 6, 7\}$ detected as *Gravel* and *Bitumen*. As for the plot of $\boldsymbol{\Psi}$, the same remarks of Fig. 11(c) about balancing the constant SVs in patches apply on Fig. 13(c), but, since there are many replicas, the figure becomes hard to read. Finally, we notice again that, for $k = 1$, the scaling factors are almost equal.

Second, in Fig. 14, we notice that MultiHU-TD is done based on spectral and morphological properties, where we observe three bundles: $\{1, 3, 6\}$, $\{2, 4, 7, 8\}$, and $\{5\}$. $\boldsymbol{B}_1$, $\boldsymbol{B}_3$, and $\boldsymbol{B}_6$ are identified as vegetation, which, respectively, reflects the areas highlighted in $\boldsymbol{A}_1$ (small vegetation areas such as trees), $\boldsymbol{A}_3$ (big vegetation areas such as meadow), and $\boldsymbol{A}_6$ (dark shadows on vegetation areas). Therefore, while $\boldsymbol{B}_1$, $\boldsymbol{B}_3$, and $\boldsymbol{B}_6$ form a bundle, unlike patches, the corresponding

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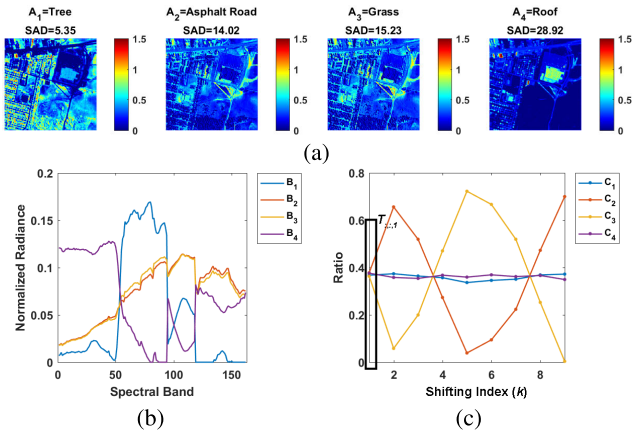


Fig. 15. Urban. CPD results of the patch-tensor for $R = 4$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$. (c) Components of $\boldsymbol{\Psi}$.

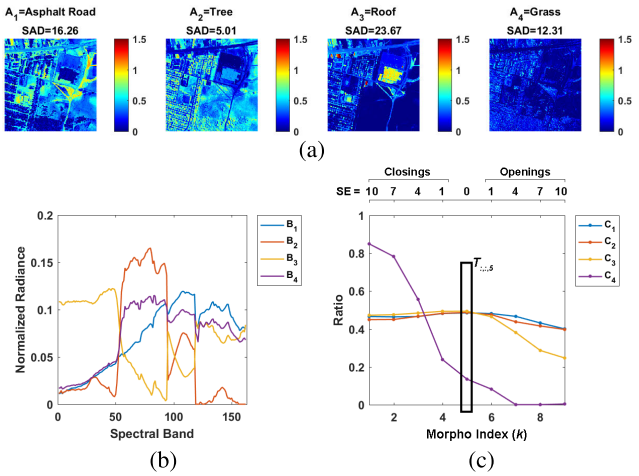


Fig. 16. Urban. CPD results of the MM-tensor for $R = 4$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$. (c) Components of $\boldsymbol{\Psi}$.

AMs highlight interesting features accounting for the scale and brightness of vegetation objects, which we discuss in more depth with the interpretation of $\boldsymbol{\Psi}$ and which applies to the other components as well. $\boldsymbol{B}_2$, $\boldsymbol{B}_4$, $\boldsymbol{B}_7$, and $\boldsymbol{B}_8$ are identified as *Bitumen* and *Gravel*, which, respectively, reflects the areas highlighted in $\boldsymbol{A}_2$ (small or short areas of roads and roofs), $\boldsymbol{A}_3$ (big connected areas of roads, roofs, and parking lots), $\boldsymbol{A}_7$ (dark shadows on parking lots and buildings), and $\boldsymbol{A}_8$ (tiny bright vehicles). Finally, $\boldsymbol{B}_4$ is identified as *Metal Sheets*, which are small.

Looking at Fig. 14(c), we observe four main patterns that can be associated with the chosen morphological parameters: First, $\boldsymbol{\Psi}_6$ and $\boldsymbol{\Psi}_7$ correspond to dark features (as observed in $\boldsymbol{A}_6$ and $\boldsymbol{A}_7$) as they have higher values when k corresponds to CbR and then continue decreasing toward ObR. Second, $\boldsymbol{\Psi}_1$, $\boldsymbol{\Psi}_2$, and $\boldsymbol{\Psi}_5$ correspond to small features (as observed in $\boldsymbol{A}_1$, $\boldsymbol{A}_2$, and $\boldsymbol{A}_4$) as they have higher values around $k = 5$ where the SEs are small. Third, $\boldsymbol{\Psi}_3$ and $\boldsymbol{\Psi}_4$ correspond to big features (as observed in $\boldsymbol{A}_3$ and $\boldsymbol{A}_4$) as they have higher values when k corresponds to big SEs with ObR. Fourth, $\boldsymbol{\Psi}_8$ corresponds to the tiny vehicles as it is the highest when k corresponds to the smallest SE. Finally, we talk about the original HSI in the following (i.e., $\boldsymbol{\mathcal{T}}_{:,:,5}$).

TABLE VI

URBAN. THE RESULTS OF AO-ADMM-ASC AND NAIVE ASC IN TERMS OF RMSE AND EXECUTION TIME: R IS THE NUMBER OF LATENT COMPONENTS. THE RESULTS OF THE MINIMUM RMSE ARE SHOWN

Algorithm	R	RMSE %	Time (s)
Naive ASC [14]	8	7.88	124
AO-ADMM-ASC	8	6.87	251

As for $k = 5$, we notice that $\boldsymbol{\Psi}_1$, $\boldsymbol{\Psi}_2$, $\boldsymbol{\Psi}_5$, and $\boldsymbol{\Psi}_8$ have the highest scaling factors (relatively bright objects including the vehicles), $\boldsymbol{\Psi}_3$ and $\boldsymbol{\Psi}_4$ have lower factors (darker objects such as asphalt roads, building roofs, parking lots, bare soil, and meadow areas), and $\boldsymbol{\Psi}_6$ and $\boldsymbol{\Psi}_7$ have the lowest factors (dark shadowy features). These relationships showcase the separability of MultiHU-TD when the third mode has a significant physical meaning and when the number of latent components is set to be sufficiently high, which can also be interpreted in terms of ELMM and balancing the SV factors.

B. Results—Urban

In this section, we present the experiments of the Urban HSI following the same order as Pavia. Since we have the same observations, and in order to avoid repetition, we briefly go over the results. However, first, we note that the spatial and spectral references already come with the downloaded dataset, consisting of four EMs, which we use as a spectral reference [as shown in Fig. 10(c)] in the experiments and four AMs [shown in Fig. 10(b)]. For the MM-HSI tensor, our SEs are disks with the successive radii: $\{1, 4, 7, 10\}$ pixels. Both Patch-HSI and MM-HSI tensors then have $K = 9$ frontal slices and dimensions $94249 \times 162 \times 9$. Finally, we also choose $R = 4$ and $R = 8$ for the number of latent components.

1) AO-ADMM-ASC: Table VI shows the RMSE results of MM-CPD between AO-ADMM-ASC and Naive ASC [14], where, again with AO-ADMM-ASC, we gain in RMSE with a small difference in the execution time.

2) Few Latent Components, ELMM, and SV: Here, we discuss the results for $R = 4$, where: 1) Fig. 15 represents Patch-CPD and 2) Fig. 16 represents MM-CPD. A fast look at the figures shows that we have the same observations as those of Pavia:

First, in Fig. 15, $\boldsymbol{B}_2$ and $\boldsymbol{B}_3$ form a bundle, and $\boldsymbol{A}_2$ and $\boldsymbol{A}_3$ are *replicas* and represent *Asphalt* + *Grass*. Moreover, in Fig. 15(c), we see the same patterns and fluctuations that were observed in Fig. 11(c) related to the constant SV and its quantitative balance in patches, and the scaling factors are equal for $k = 1$. On the other hand, $\{\boldsymbol{A}_1, \boldsymbol{B}_1\}$ and $\{\boldsymbol{A}_4, \boldsymbol{B}_4\}$ represent *Tree* + *Grass* and *Roof*, respectively, with steady $\boldsymbol{\Psi}_1$ and $\boldsymbol{\Psi}_4$ patterns.

Second, in Fig. 16, while $\boldsymbol{B}_4$ and $\boldsymbol{B}_1$ form a bundle, we notice that $\boldsymbol{\Psi}_4$ has the same pattern observed in Fig. 12(c), which corresponds to dark shadows and is reflected in $\boldsymbol{A}_4$, which highlights shadows of buildings and trees that fall on grass areas. As for the other components, they can be interpreted similar to those in the case of Pavia (including for $k = 5$), where $\{\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{\Psi}_1\}$, $\{\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{\Psi}_2\}$, and $\{\boldsymbol{A}_3, \boldsymbol{B}_3, \boldsymbol{\Psi}_3\}$

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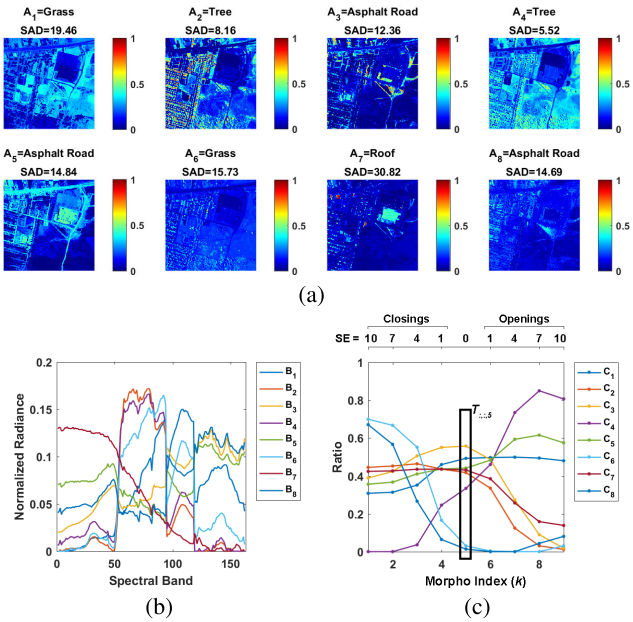


Fig. 17. Urban. CPD results of the MM-tensor for $R = 8$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$. (c) Components of $\boldsymbol{\Psi}$.

represent *Asphalt* + *Grass*, *Tree* + *Grass*, and *Roof*, respectively.

3) More Latent Components, ELMM, and Multifeature Separability: Here, we discuss the results for $R = 8$, where Fig. 17 represents MM-CPD. We skip the case of Patch-CPD in order to avoid repetition, where we simply end up with more *replicas* of the components of Fig. 15(a) and (b). In Fig. 17, we are interested in the features of the AMs that do not appear in Patch-CPD as the comments on the spectral and morphological patterns are the same as those of Pavia; where the plots reflect the qualitative features that appear in the respective AMs.

We observe three column sets forming three spectral bundles: $\{1, 2, 6\}$, $\{3, 5, 7, 8\}$, and $\{4\}$. First, $\boldsymbol{A}_1$, $\boldsymbol{A}_2$, $\boldsymbol{A}_4$, and $\boldsymbol{A}_6$ were identified as *Vegetation*: $\boldsymbol{A}_1$ highlights grass fields, which is close to the *Grass* reference and does not appear in Patch-CPD. $\boldsymbol{A}_2$ and $\boldsymbol{A}_4$ highlight small and big areas, and together they correspond to the *Tree* reference. $\boldsymbol{A}_6$ highlights dark shadows (which is reflected in $\boldsymbol{\Psi}_6$). Second, $\boldsymbol{A}_3$, $\boldsymbol{A}_5$, and $\boldsymbol{A}_8$ are identified as *Asphalt Road*: $\boldsymbol{A}_8$ seems to correspond to dark features (refer to $\boldsymbol{\Psi}_8$). $\boldsymbol{A}_3$ highlights small roads such as dirt and narrow streets, while $\boldsymbol{A}_5$ highlights large roads, such as the main and connected roads, which are clearly highlighted, unlike the case of patches. Third, $\boldsymbol{A}_7$ is identified as *Roof* and highlights both small and large building roofs.

VI. Conclusion

In this article, we proposed a methodological framework for MultiHU-TD based on CPD and the AO-ADMM-ASC algorithm, where the samples (pixels) represent a convex combination of the sources. We also established a unified framework for the interpretability of MultiHU-TD into "multilinear" subspaces that involved mathematical, physical, and graphical representations of the CPD model with ASC, ELMM, and SV. Finally, we proposed to include MM as spatial features in a spectral–spatial HU and dived further into the case of neighborhood patches, where MM incorporates physically meaningful features into the data tensor. Through the comparison between the two third-mode examples, we provided in-depth insights on the interpretability of MultiHU-TD including the physical significance of the factor matrices and the input rank. To conclude, we summarize some key properties of MultiHU-TD as follows.

1) Multifeature hyperspectral data are useful for low-rank latent variable analysis, such as unmixing.
2) Having multiple modalities of features allows to exploit more information on the scene, relaxing the dependence on the high-rank spatial structures while conserving enough context of the scene.
3) Having multiple modalities of features with CPD acts as an implicit prior on the scene. The MultiHU-TD framework is then equivalent to performing a coupled matrix decomposition on each of the tensor slices where the abundances' matrix $\boldsymbol{A}$ is the common factor.
4) Multifeature unmixing through low-rank tensor decomposition factorizes the pixel and spectral information and implicitly models the spectral variabilities of the scene.

In the future, we plan to explore BTD that allows some flexibility with the tensor structure and can be seen as an extension to spectral bundles for SV [11] but also comes with many challenges such as the rank and the interpretation of the subspaces. Moreover, areas of BSS other than HU may be explored. Finally, it is worth mentioning that some deep learning approaches are being considered for HU (which still suffer from the increasing and flexible dimensionality of HSIs and the difficulty of finding datasets for training, especially in a blind framework). However, by developing our methodological study of tensor-based unmixing and pushing for interpretability, this framework can help interpretability in data-driven methods based on tensor decomposition [21], [58], [59].

Appendix A
Sparse-NMF Results on the Matricized HSI

In this appendix, we include the results obtained by applying sparse NMF (with ASC) [17], which partly inspired this work. We note that these results cannot be compared with those of tensor decomposition in terms of RMSE and execution time due to the following reasons.

1) In terms of RMSE, on the one hand, we reconstruct a multifeature HSI tensor, while, on the other hand, we reconstruct an HSI matrix that does not apply in *multifeature analysis*, so the reconstructed data represent different types of information.
2) In terms of execution time, NMF typically has shorter execution times than tensor-based methods due to the added complexity. However, both tools are fundamentally different and cannot be used for the same multilinear application.

With that said, the obtained results serve only as a qualitative baseline or reference for the AMs and *spectral components* of decomposing the two HSI datasets, which can indeed be

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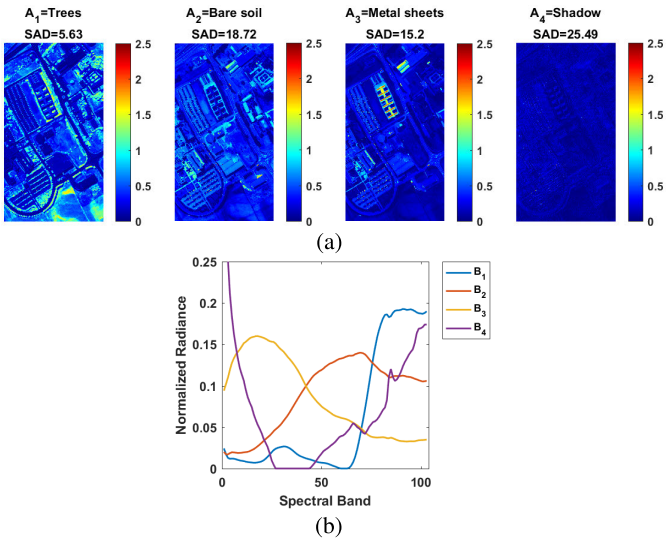


Fig. 18. Pavia. NMF results of the HSI matrix for $R = 4$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$.

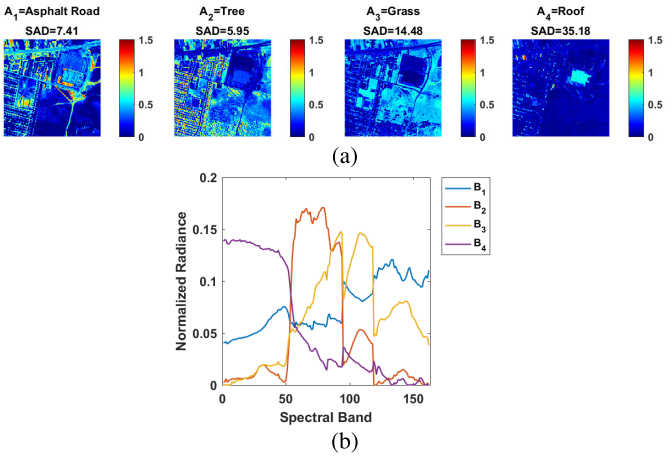


Fig. 19. Urban. NMF results of the HSI matrix for $R = 4$. (a) Components of $\boldsymbol{A}$. (b) Components of $\boldsymbol{B}$.

compared with those obtained in the case of MM-HSI and Patch-HSI tensors.

A. Pavia Dataset

Fig. 18 shows the results obtained for the dataset of Pavia University. Each AM is shown with the class that is assigned to it based on the minimum SAD value, which is reported as well.

First, we look at the first three components: *Trees*, *Bare Soil*, and *Metal Sheets*. Their spectral signatures $\{\boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{B}_3\}$ shown in Fig. 15(b) look very similar to those of the reference, but the corresponding SAD values are relatively bad (high) compared to those obtained using tensor decomposition in Figs. 11 and 12.

The AMs $\{\boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{A}_3\}$ of Fig. 18(a) show highlighted elements belonging to *Trees*, *Bare Soil*, and *Metal Sheets*, respectively. However, we notice that other areas of the scene belonging to these categories are barely or faintly highlighted (e.g., asphalt roads, brick parking lots, and other soil areas), which is due to the insufficiency of LMM to model their variabilities.

Regarding the *Shadow* component, the spectral signature $\boldsymbol{B}_4$ looks slightly similar to the reference, but the shadows in the map are barely visible due to their relatively very low brightness $\boldsymbol{A}_4$.

TABLE VII
SPECTRAL MIXTURE OF EACH OBJECT BASED ON THE EMS, STREET, VEGETATION, AND METAL SHEETS

Object number	1	2	3	4	5	6
Street (%)	10	0	0	**80**	20	**100**
Vegetation (%)	**70**	**100**	0	10	20	0
Metal Sheets (%)	20	0	**100**	10	**60**	0
Total (%)	100	100	100	100	100	100

B. Urban Dataset

Fig. 19 shows the AMs and spectral signatures of $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively. We obtain four components with relatively low (good) SAD values and good AMs similarity with respect to the reference. Finally, we also note that some dark areas such as asphalt roads are not highlighted.

APPENDIX B
SYNTHETIC HSI EXAMPLE

In the following, we simulate a time-series HSI tensor through a synthetic example. The noiseless tensor is reconstructed from its building blocks, i.e., the factor matrices, which are considered here as the noiseless GTs. This simplified example serves as a demonstration of the performance of MultiHU-TD as we increase the SNR.

1) Description of the Synthetic Dataset: The synthetic HSI cube $\boldsymbol{\mathcal{M}}$ of a signle time-stamp has dimensions $128 \times 128 \times 26$, whose matricized version is denoted by $\boldsymbol{M}$ of dimensions 16384×26. In particular, the following holds.

1) The spatial scene is composed of six objects that vary in size, as shown in Fig. 20, denoted by "Object 1" through "Object 6."
2) Three independent spectral signatures are simulated from the HSI of Pavia University, corresponding to `Street`, `Vegetation`, and `Metal Sheets`, which are shown in Fig. 21(b).
3) The objects are assigned linear mixtures of the three simulated spectra, as shown in Table VII. This means that the GT of the abundance satisfies the ASC.

That said, $\boldsymbol{M}$ has rank 3 where $\boldsymbol{M} = \boldsymbol{A}\boldsymbol{B}^{\mathsf{T}}$ such that $\boldsymbol{A} \in \mathbb{R}^{16384\times 3}$ and $\boldsymbol{B} \in \mathbb{R}^{26\times 3}$ are two factor matrices. The spatial and spectral GTs, of $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively, are shown in Fig. 21(a) and (b).

As for the full time-series HSI tensor, it is composed of three stamps where the objects of the scene change in color or disappear in time. In principle, this corresponds to a time-series tensor $\boldsymbol{\mathcal{D}}$ of dimensions $128 \times 128 \times 26 \times 3$. After reordering the pixel modalities in lexicographic order, we would obtain a tensor $\boldsymbol{\mathcal{T}}$ of dimensions $16384 \times 26 \times 3$. We synthesize the time-series HSI tensor $\boldsymbol{\mathcal{T}}$ of dimensions

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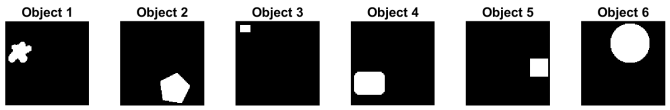


Fig. 20. Simulated spatial objects of the synthetic HSI, denoted as Objects 1–6.

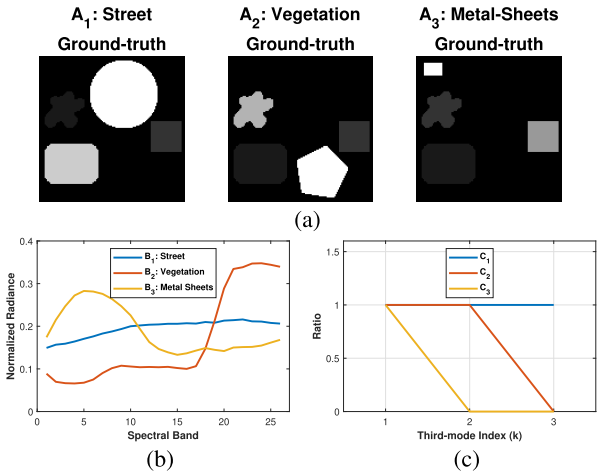


Fig. 21. GT components of the synthetic tensor $\mathcal{T}$. (a) GT of $\boldsymbol{A}$. (b) GT of $\boldsymbol{B}$. (c) GT of $\boldsymbol{C}$.

$\mathbb{R}^{16384\times 26\times 3}$ from the product with an additional matrix $\boldsymbol{C} \in \mathbb{R}^{3\times 3}$ such that

$$\mathcal{T} = \mathcal{I} \underset{1}{\bullet} \boldsymbol{A} \underset{2}{\bullet} \boldsymbol{B} \underset{3}{\bullet} \boldsymbol{C} \tag{32}$$

where $\mathcal{I} \in \mathbb{R}^{3\times 3\times 3}$ is a diagonal tensor of ones and $\boldsymbol{C}$ is described as follows:

$$\boldsymbol{C} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} <=> \begin{pmatrix} k=1 \\ k=2 \\ k=3 \end{pmatrix} \tag{33}$$

where k is the index spanning the third modality, which is that of time stamps. C is considered the GT of the temporal signatures, and its columns are plotted in Fig. 21(c).

In order to assess the performance of the proposed framework in the presence of noise, Gaussian noise was added on the tensor $\mathcal{D}$ with varying levels of noise, where the variance $\sigma^2 \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, as described in Fig. 22. After adding noise, the tensor is reshaped back to $\mathcal{T}$.

2) Results of Unmixing Without Adding Noise: Fig. 23 shows the components of the factor matrices after decomposing the synthetic time-series HSI tensor described using CP decomposition. The components of the factor matrices are perfectly recovered thanks to the CP uniqueness of the data tensor [60]. Moreover, it is worth noting that the RMSE between the original data tensor and the reconstructed one is 0, which means perfect reconstruction.

3) Results of Unmixing With Varying Levels Of Noise: In this case, we discard the reconstructability of the tensor itself and focus on the factor matrices, of which we have the abundance matrix $\boldsymbol{A}$. Figs. 24–27 show the results of decomposing the tensor under varying levels of Gaussian noise, with variances of 10^{-4}, 10^{-3}, 10^{-2}, and 10^{-1}, respectively. Moreover, Fig. 28 shows the evolution of the SAD index of the

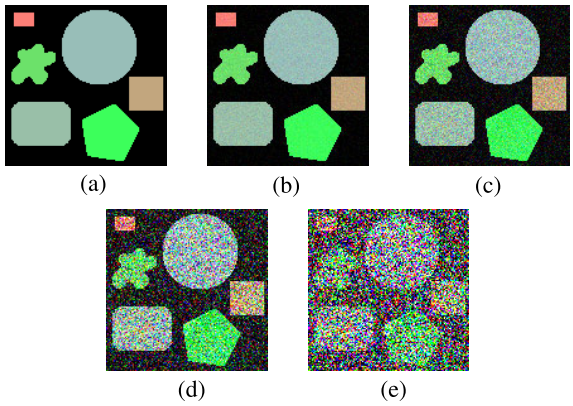


Fig. 22. Synthetic HSI scene at a single time stamp $k = 1$ in false colors with Gaussian noise of variance σ^2 ranging from 0 (being noiseless) to 10^{-1} (being very noisy). (a) Noiseless. (b) $\sigma^2 = 10^{-4}$. (c) $\sigma^2 = 10^{-3}$. (d) $\sigma^2 = 10^{-2}$. (e) $\sigma^2 = 10^{-1}$.

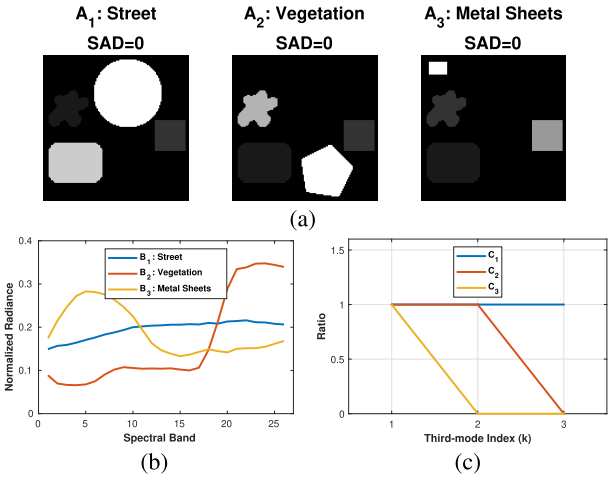


Fig. 23. Synthetic HSI. Results of decomposing the HSI tensor $\mathcal{T}$ without noise, where we also have RMSE = 0. (a) Recovered columns of $\boldsymbol{A}$. (b) Recovered columns of $\boldsymbol{B}$. (c) Recovered columns of $\boldsymbol{C}$.

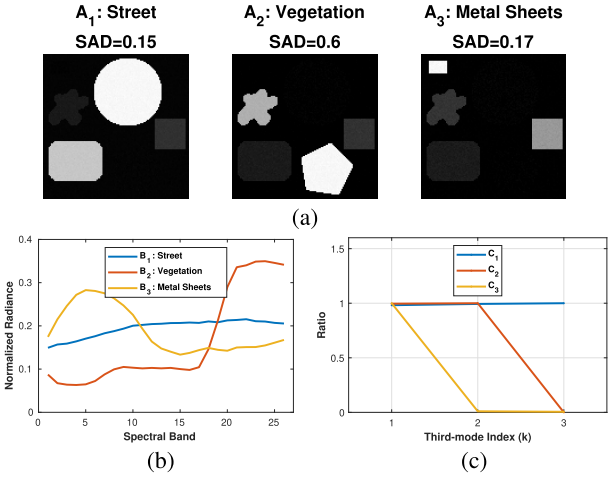


Fig. 24. Synthetic HSI. Results of decomposing the HSI tensor $\mathcal{T}$ with Gaussian noise of variance 10^{-4}. (a) Recovered columns of $\boldsymbol{A}$. (b) Recovered columns of $\boldsymbol{B}$. (c) Recovered columns of $\boldsymbol{C}$.

estimated EMs as the noise level increases. We note that the experiments are done without applying any spatial denoising.

The components of the factor matrices, including the abundance matrix $\boldsymbol{A}$ whose rows sum to one, are quite recoverable

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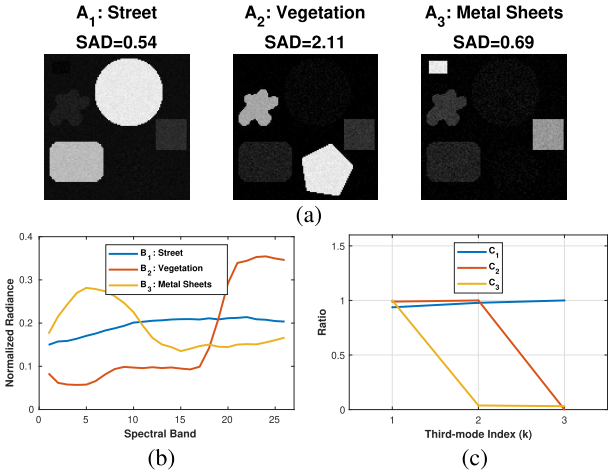


Fig. 25. Synthetic HSI. Results of decomposing the HSI tensor $\mathcal{T}$ with Gaussian noise of variance 10^{-3}. (a) Recovered columns of $\boldsymbol{A}$. (b) Recovered columns of $\boldsymbol{B}$. (c) Recovered columns of $\boldsymbol{C}$.

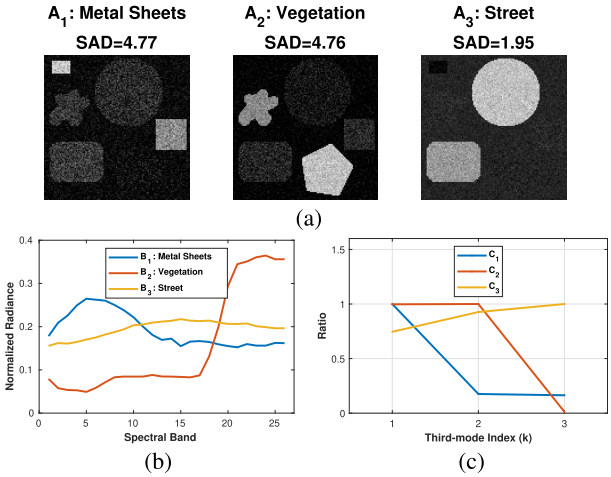


Fig. 26. Synthetic HSI. Results of decomposing the HSI tensor $\mathcal{T}$ with Gaussian noise of variance 10^{-2}. (a) Recovered columns of $\boldsymbol{A}$. (b) Recovered columns of $\boldsymbol{B}$. (c) Recovered columns of $\boldsymbol{C}$.

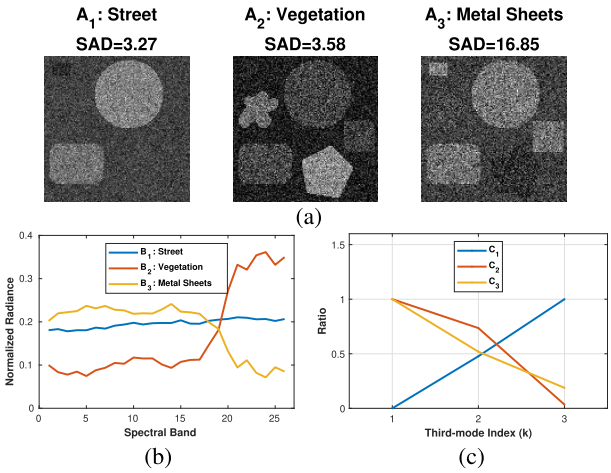


Fig. 27. Synthetic HSI. Results of decomposing the HSI tensor $\mathcal{T}$ with Gaussian noise of variance 10^{-1}. (a) Recovered columns of $\boldsymbol{A}$. (b) Recovered columns of $\boldsymbol{B}$. (c) Recovered columns of $\boldsymbol{C}$.

up to a level of noise of variance $\sigma^2 = 10^{-3}$. In the case of $\sigma^2 = 10^{-2}$, the components are still recoverable even though the tensor looks quite noisy in Fig. 22(d). In the case of

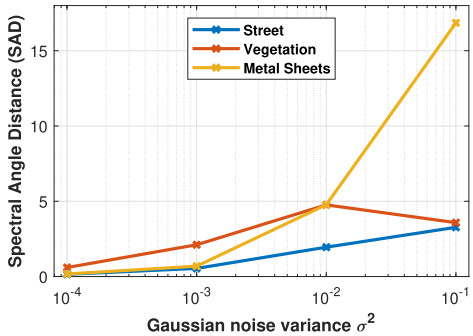


Fig. 28. Synthetic HSI. Evolution of the SAD index of the estimated spectra compared to the spectral GT.

$\sigma^2 = 10^{-1}$, which is very noisy that some of the objects are indistinguishable in Fig. 22(e), the reconstructed spectra are still fairly close to their GTs, and the spatial structures in the factors of the abundance matrix can still be recognized. In Fig. 28, we can see that the SAD index generally increases with the level of noise. However, the values remain quite small, i.e., less than 5°, indicating the recoverability of the estimated EMs even under such high levels of noise. The only exception in this case is that of *Metal Sheets* under a variance of 10^{-1}, which results in an SAD index of about 16°; this can be due to the small size of *Metal Sheets* objects, which makes them more susceptible to noise.

Moreover, we note that, with the addition of noise, the tensor $\mathcal{T}$ that was synthesized to be of rank $R = 3$ becomes full rank. Decomposing the noisy tensor with a low rank can be roughly seen as a denoising procedure since it forces the projection of the data onto a lower rank multilinear latent subspace of rank $R = 3$. However, spatial denoising is still needed in order to recover a better representation of the spatial components, but said application is out of the scope of this article.

Finally, this is a simple, controlled, and minimalistic example that serves as an intuition for more complex structures where the situation is completely blind, such as in real HSI tensors.

Appendix C
Computational Complexity of AO-ADMM-ASC

First, we refer to paper [48] for detailed explanations concerning the CPD by using AO-ADMM. Let us consider a third-order tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ (where $I_1 \ggg I_2 I_3$) with a low rank R, and we consider the complexity as per ADMM iterations. Also, it is important to note that matrices $\tilde{\boldsymbol{W}}$ and $\boldsymbol{T}$ are independent of the inner ADMM updates, so they can be used only once to compute the products $\tilde{\boldsymbol{W}}^\top \tilde{\boldsymbol{W}}$ and $\tilde{\boldsymbol{W}}^\top \boldsymbol{T}$, whose values can be cached before the ADMM update allowing to save a lot of repetitive computations. Now, we split the problem into three steps.

1) *Unconstrained CPD:* In this case, the complexity of the algorithm is dominated only by the updates of the factor matrices. Hence, the complexity is $\mathcal{O}(I_d R^2)$ $\forall d \in \{1, 2, 3\}$.
2) *Nonnegative CPD That Is Relevant for the Updates of Each of the Factor Matrices:* Nonnegativity requires

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only elementwise projection, i.e., a complexity of $\mathcal{O}(I_d R)$ $\forall d \in \{1, 2, 3\}$, which is negligible compared to $\mathcal{O}(I_d R^2)$. Hence, the complexity is still dominated by $\mathcal{O}(I_d R^2)$.

3) *Nonnegative CPD With Sparsity and ASC, Which Is Only Relevant for the Update of* $\mathbf{A}$*: Sparsity* is like nonnegativity as it boils down to an elementwise subtraction with complexity $\mathcal{O}(I_1 R)$, which is negligible compared to $\mathcal{O}(I_1 R^2)$. As for ASC, it requires two updates.
   a) $b_{J+1,r} = \delta \psi_{K,r}^{-1}$ $\forall r \in \{1, \ldots, R\}$, i.e., a complexity of $\mathcal{O}(R)$, which is negligible.
   b) $t_{i,J+1,k} = \sum_{r=1}^{R} a_{i,r} b_{J+1,r} \psi_{k,r}$ $\forall i \in \{1, \ldots, I\}$ and $\forall k \in \{1, \ldots, K-1\}$, i.e., a complexity of $\mathcal{O}(I_1 I_3 R)$.

Considering that MultiHU-TD admits a *low-rank* CP decomposition, R is usually small, and in most of the cases, we would have $R < I_3$ (or at least very close). In this case, the complexity is dominated by $\mathcal{O}(I_1 I_3 R)$, which changes linearly with either the number of pixels I_1, the third-mode features I_3, or the latent components R.

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